

6 (a) (i) State what is meant by *transpiration*.

.....

.....

.....[3]

(ii) Explain how the transpiration stream is important for the production of proteins in a plant.

.....

.....

.....[2]

(b) Describe and explain how tissues in the stem of a plant become stained when a plant is carefully removed from the soil and placed in a solution of dye for a few hours.

.....

.....

.....

.....

.....

.....

.....[5]

[Total: 10]

7 (a) Describe the functions of the kidneys.





6 (a) (i) loss of water;
as vapour / evaporation;
from leaves / aerial parts of a plant / to the atmosphere;
through stomata (A between or through guard cells);

[max. 3]

(ii) brings nitrates;
dissolved in water;
from soil / to combine with carbohydrates;

[max. 2]

(b) dye diffuses (R ref. to osmosis);
through permeable (R semi- / partially- / selectively-);
(Ignore refs. to cell membrane)
(cell) wall;

of root (hair) cells;

xylem (any correct ref.);

ref. to travelling upwards (A refs. to transpiration pull / stream capillarity / root pressure AW);

[max. 5]

[Total: 10]

7 (a) Describe the functions of the kidneys.



.....

.....

.....

.....

.....

.....

.....[4]

- 7 (a) excretion (A excretory);
removal of urea ;
salts / ref. osmoregulation / varying amounts of water;
toxins / poisons / hormones (R harmful);
filtration of / removal from + blood / plasma;
forming urine;

[max. 4]

- (b) ref dialysis;
blood along a partially permeable tube;
washing / bathing fluid (dialysis fluid could score 2 marks);
ref controlled contents of bathing fluid (i.e. concentrations as required in blood)
(Look for more than just a ref. to urea.);
diffusion (any correct ref.);
keeps blood at constant osmotic potential;
washing fluid changed;
temperature regulation;
urea removed;
blood from and to patient ;
kidney transplants / donation;
bubble trap / adding anticoagulants / counter flow / large surface area / pump ;
(A coiled tube)

[max. 6]

[Total: 10]

..... [2]

. [2]

[8]

[Total: 10]

7 (a) (i) State **one similarity** and **one difference** in the functions of the urethra in

paC

6 (a) transport sucrose / sugar / amino acids (R glucose / food);
in solution;
ref. direction of movement (A around the plant);

[2]

(b) osmosis;
through cell membrane;
of root hair;
ref. movement from cell to cell;
ref. movement through or between cell walls;
enters xylem;
transpiration pull / stream / capillarity / molecular cohesion / root pressure;
evaporation / ref. water vapour;
from mesophyll cells / into air spaces;
ref. water potential gradient;
diffusion;
through stomata;

[8]

[Total: 10]

7 (a) (i) State **one similarity** and **one difference** in the functions of the urethra in
and in a female adult person.

similarity

.....

difference

..... [2]

- 7

(a)

(i)

carries urine (R just urea) in male and female;
carries semen / sperms / gametes AW in male;

[2]

(ii)

ref. size comparison;
ref. qualified numbers comparison;
ref. mobility comparison;

[3]
- (b)

(advantages)

only needs to be undergone once AW (e.g. less trouble);
high reliability / effectiveness AW;

[2]

(disadvantages)

does not protect against sexually transmitted diseases;
ref. risks of surgery / anaesthetic;
not a temporary solution;
difficult / expensive to reverse;
need access to medical services AW;

[4 max]

[Total: 10]

(b) List three structural changes that must occur in young, unmodified plant cells as they develop into xylem tissue.

- 1
- 2
- 3 [3]

(c) A small, leafy branch is cut from a tree. After some hours, the stem of the branch remains firm but the leaves become limp. Suggest an explanation for this.

stem remains firm

.....

.....

leaves become limp

.....

.....

..... [5]

[Total: 11]

4 Fig 4.1 shows a section through a ripe fruit of a tomato plant



- 3 (a) (i) cell wall + correct drawing (outside existing line);
- (ii) nucleus + correct drawing (must be in cytoplasm);
- (iii) vacuole / cytoplasm + correct drawing (if vacuole, must be larger than the nucleus.); [3]

If no marks scored through unacceptable drawings, allow max. 1 if all 3 are shown in the correct positions – vacuole in the middle, cell wall on the outside and nucleus between.

(b) elongation of cell (R cylindrical);

loss of cell contents / or one named content (A ref. death of cell / hollow)
(R chloroplasts / dissolving of cell contents.);

loss of end walls;

addition of thickening / strengthening material / lignin; [max. 3]

(c) (*stem*) contains xylem;

(xylem is) thickened / strengthened / contains lignin;

provides support;

(*leaf*) xylem less important for support;

loss of water / transpiration;

water not replaced / doesn't reach leaves;

cells become flaccid / lose turgor AW (R plasmolysed);

cells unable to support weight of leaf / wilts; [max. 5]

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[8]

[2]

[Total: 10]

9 (a) toxic / poisonous materials (R harmful);

waste products + metabolism AW;
(R refs. to salts)

urea;

liver / blood + kidneys;

urine;

urethra;

sweat;

skin / (epi)dermis;

carbon dioxide;

from cells / blood + alveoli;

exhaled / breathed out / expelled or removed from lungs AW; [max. 8]

(b) undigested food;

cellulose / fibre / lignin / roughage;

not from metabolism AW; [max. 2]

1 Fig. 1.1 shows a plant with a magnified transverse section (TS) of the root and an incomplete, magnified transverse section (TS) of the stem.

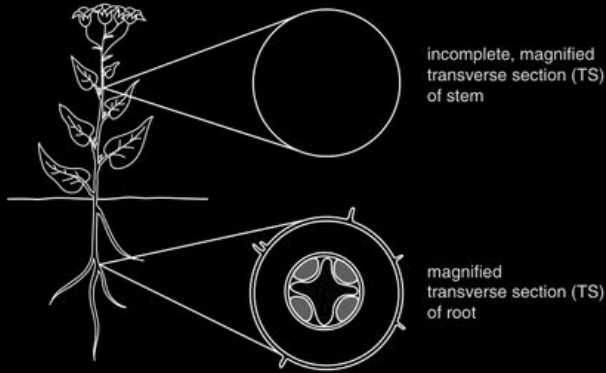


Fig. 1.1

- (a) The transverse section of the root in Fig. 1.1 shows the distribution of xylem and phloem tissue.
- (i) Using a labelling line and the letter **X**, show, on Fig. 1.1, the location of the xylem tissue in the transverse section (TS) of the root. [1]
 - (ii) Complete the transverse section (TS) of the stem in Fig. 1.1 to show the distribution of xylem and phloem tissue. [2]
 - (iii) Using a label line and the letter **P**, show the location of the phloem tissue that you have drawn. [1]

(b) State the functions of phloem tissue.

.....

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.....

.....

.....

..... [3]

[Total: 7]

1 (a) (i)	Label line X on xylem on transverse section of root ;		[1]
(ii)	vascular bundle drawn towards outside of transverse section of stem ; vascular bundle divided into two sections ; three or more vascular bundles drawn ;		[max 2]
(iii)	label line P drawn in correct location (outer side of vascular bundle) ;		[1]
(b)	translocation ; transport ; dissolved / in solution ; sugar(s) / sucrose / products of photosynthesis ; amino acids ; reference from source + to sink / leaves to roots AW ;		[max 3]
[Total: 7]			

2 A potometer is used to measure water uptake by a plant. Fig. 2.1 shows the stem and flower of a plant in a potometer. As water is taken up, the bubble moves in the direction shown.

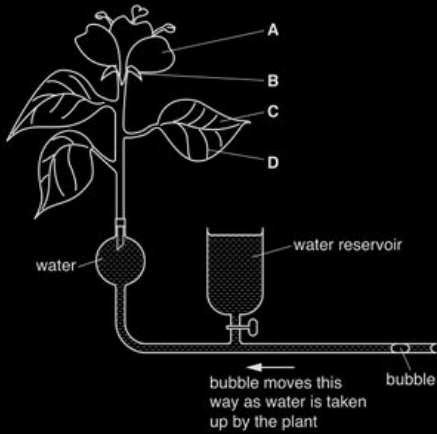
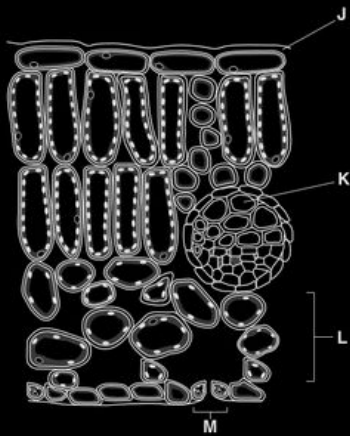


Fig. 2.1

- (a) Name the parts **A**, **B**, **C** and **D** in Fig. 2.1.
- A**
- B**
- C**
- D** [4]
- (b) Describe the pathway taken by water as it moves from the potometer, through the plant stem and into the surrounding air.
-
-
-
-
- [3]
- (c) (i) In an experiment, the bubble moved a distance of 60 mm in 10 minutes.
Calculate the average rate at which the bubble moved in mm per min.
- mm per min [1]
- (ii) The experiment was repeated in an area of lower light intensity.
- Predict and explain what would happen to the rate at which the bubble moved.
- prediction.....
-
- explanation.....
-
- [3]
- [Total: 11]

2(a)	(A) petal ; (B) sepal ; (C) leaf / lamina / cuticle ; (D) vein / midrib / vascular (tissue) OR xylem + phloem ;	4	
2(b)	xylem ; to leaves / flower(s) ; between leaf cells ; surface of mesophyll cells ; (leaf) air spaces ; (through) stomata / guard cells ;	3	
2(c)(i)	6 ;	1	
2(c)(ii)	reduces / lower(s) AW ; less photosynthesis ; less water needed ; stomata / guard cells + open less / are closed ; less diffusion ; less evaporation / transpiration ;	3	
	Total:	11	

6 The diagram shows a magnified transverse section through a leaf.



Name each of J, K, L and M and explain the importance of each in the process of transpiration.

J

K

L

M

[10]

[Total: 10]

6	(J) <u>cuticle</u> ; waterproof / waxy AW ; prevents / reduces + water loss / transpiration / evaporation ; (K) <u>xylem</u> ; transport + water ; from named location to named higher location ; (L) <u>spongy</u> ; air / intercellular + spaces ; water film AW ; evaporation / water vapour ; diffusion ; (M) stoma / stomata ; guard cell ; open / close ; reference to water <u>vapour</u> + exits ; diffusion ;	10	Note marking points must not be transferred between the sections R 'phloem' for xylem
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3 The photograph shows an insect called a leafhopper. When it feeds, its mouthparts penetrate the xylem of the plant on which it lives.



Leafhoppers carry disease-causing bacteria from plant to plant.

- (a) (i) State the term used for a disease-causing organism.
..... [1]
- (ii) State the term used for an animal, such as an insect, that carries disease-causing organisms.
..... [1]
- (b) The bacteria carried by the leafhopper reproduce inside the plant stem and block the xylem.
- (i) Suggest and explain why this can cause the leaves of the plant to wilt.
.....
.....
.....
..... [4]
- (ii) The leaves of plants affected by these bacteria may also develop white patches. Suggest reasons for this.
.....
.....
..... [2]
- (iii) Suggest an explanation for the fact that the fruits of plants affected by these bacteria are small and poorly developed.
.....
.....
.....
.....
..... [4]

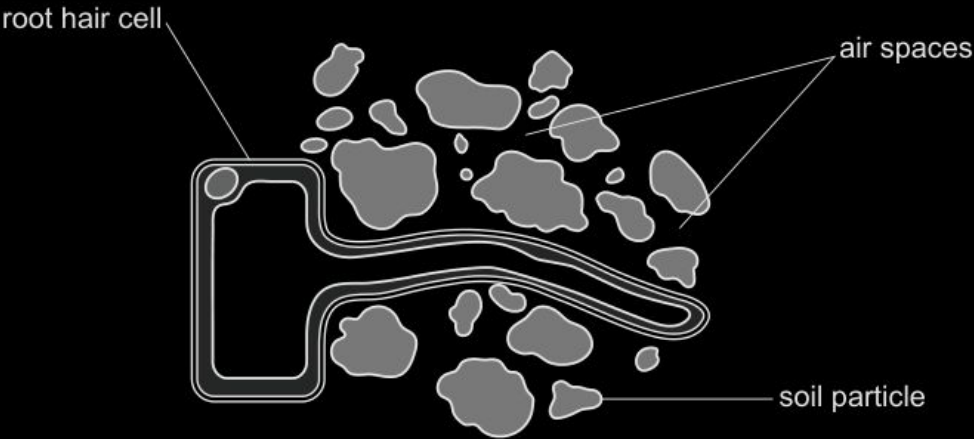
[Total: 12]

4 (a) Explain the importance of the production of carbon dioxide in bread-making.

3(a)(i)	<u>pathogen</u> ;	1
3(a)(ii)	<u>vector</u> ;	1
3(b)(i)	water lost from leaves ; by evaporation / transpiration ; cannot be replaced / cannot pass up stem ; water lost from cells ; lose their turgor / turgidity ;	4
3(b)(ii)	can't make chlorophyll ; reference to magnesium / zinc not getting to leaves ;	2
3(b)(iii)	reduced photosynthesis ; less sugar / starch / protein made ; less water available to fruit ; fewer nutrients or named nutrient sent to fruit ; less growth or development (linked to protein) ;	4

Question	Answer	Marks
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1 The diagram shows a root hair cell from a lemon tree and some soil particles.



(a) (i) State **one** feature, visible in the diagram, which shows that a root hair cell is a plant cell.

..... [1]

(ii) Describe how mineral ions pass from the soil into the root hair cell.

.....
.....
.....
.....
.....
..... [4]

(b) The lemon tree has grown in the same soil for many years so the soil now lacks magnesium.
Describe and explain the effect on the lemon tree of a lack of magnesium.

.....
.....
..... [2]

[Total: 7]

2 A person visits an eye doctor to have an eye test. On the wall of the doctor's room is a diagram of

1(a)(i)	cell wall or large / central / sap + vacuole ;	1	
1(a)(ii)	<i>any four from:</i> dissolved in water ; through the cell wall ; <u>carriers</u> in membrane ; through the cell / partially permeable AW + membrane ; by active transport / diffusion ; correct reference to concentration gradient ; using energy (for active transport) ; from (aerobic) respiration (for active transport) ;	4	
1(b)	yellowing / chlorosis / reduced growth / cannot photosynthesise / make glucose / starch / sugar / carbohydrate ; deficiency of chlorophyll ;	2	

Section C

7(a)	<i>any seven from:</i> transported in blood / plasma ; dissolved ; hepatic vein to vena cava / heart ; renal artery (to kidney) ; to kidney ; urea removed / filtered + from blood ; part of urine ; <u>ureter</u> (to bladder) ; bladder ; (bladder to) <u>urethra</u> ;	7	
7(b)	<i>any three from:</i> cycling / replenishing nitrogen ; used / converted by nitrifying bacteria / nitrogen fixing bacteria ; form + nitrates ; nitrates + taken up by plants ; production of proteins / amino acids ; <u>increased</u> soil fertility (for crops) / plants grow faster / bigger / AW ; animals / herbivores obtain protein / nitrogen from plants ;	3	



- State **two** characteristics of a fungus.

- Explain how the response of the tree to infection leads to these symptoms of disease.**

- Describe what is meant by the term **vector**.

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- Diagram illustrating the process of vegetative propagation in a tree. The roots of the parent tree grow stems called suckers. Each sucker forms a new tree.

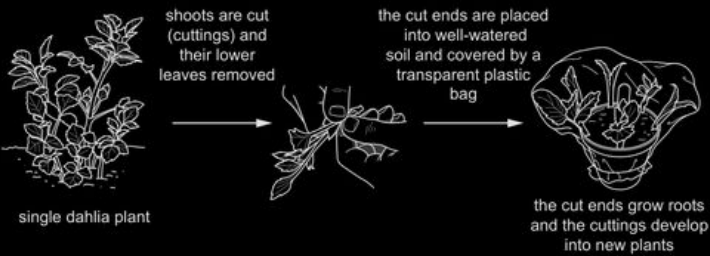
4 Cardiopulmonary resuscitation (CPR) is a first aid procedure.

3(a)	1 hyphae / mycelium ; 2 saprotrophic / saprophytic / feeds on dead matter AW / decomposer ; 3 spores ; 4 cell wall + chitin ; 5 more than one nucleus / multinucleate ;	2	
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3(b)	1 less / no + water transported / water absorbed / transpiration ; 2 water loss greater than water uptake ; 3 less + water in cells ; 4 flaccid / loss of turgor / plasmolysed ; 5 magnesium / Mg ; 6 less / no + chlorophyll ;	5	
3(c)	1 transfer / carries / spreads AW + pathogen / fungus / parasite ; 2 one organism AW to another ; 3 not itself + affected / infected ;	2	
3(d)(i)	<u>asexual</u> ;	1	
3(d)(ii)	1 close together ; 2 connected AW ; 3 genetically identical / all genetically susceptible / clones ;	3	

Question	Answer	Marks	Guidance
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3 A gardener grows a species of dahlia, a flowering plant.
Each year the gardener chooses his best plants and takes cuttings to make new plants.



(a) (i) Removing the lower leaves and placing the cuttings in a transparent bag helps prevent them wilting until the roots develop.

Suggest and explain how these two steps help to prevent wilting.

removal of lower leaves

.....

.....

covering with transparent bag

.....

.....

[3]

(ii) Taking cuttings is a commercial application of asexual reproduction.

Define asexual reproduction.

.....

..... [2]

(b) The stems of the dahlia cuttings contain xylem vessels.

Describe **two** functions of xylem vessels in young stems of dahlias.
For each function explain **one** way in which xylem structure is adapted for this function.

function 1

.....

adaptation

.....

function 2

.....

adaptation

..... [4]

(c) The gardener's dahlia cuttings grow into full-sized plants with flowers.

The gardener decides that he would like to produce a plant that combines the disease-resistant characteristics of one of these plants with the red petal colour of another.

With reference to named parts of the flowers, describe how he could achieve this.

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..... [5]

[Total: 14]

3(a)(i)	removal of leaves: reduction + surface area / stomata ;	1	
	covering with transparent bag: increase humidity ;	1	
	to reduce transpiration OR evaporation / water loss ;	1	
3(a)(ii)	any two from: production of <u>genetically identical</u> offspring / <u>clones</u> ; mitosis ; from one parent/ AW ;	2	

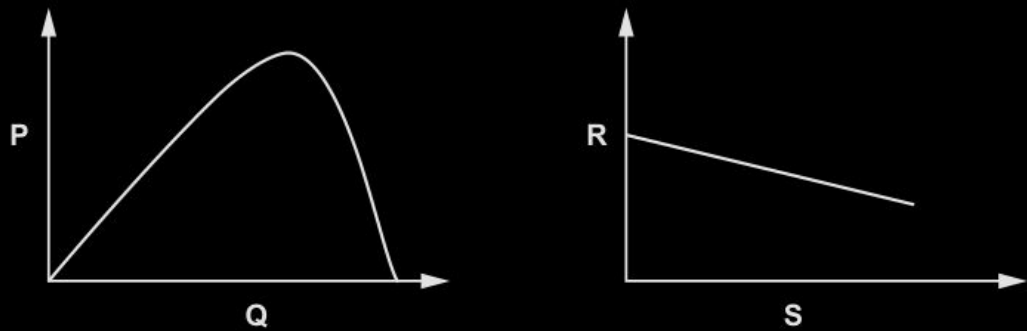
3(b)	function: xylem vessels transport water and/or minerals ;	1	
	adaptation: long / thin / no end cell walls / hollow tubes / no cytoplasm ;	1	
	function: (mechanical) support / keeps stem upright ;	1	
	adaptation: lignified OR woody / strengthened / hard + walls ;	1	
3(c)	any five from: 1 <u>artificial selection</u> / <u>selective breeding</u> ; 2 transfer pollen / pollination ; 3 from anther / stamen of one plant + e.g. red petal colour ; 4 to stigma / carpel / pistil of another ; 5 fertilisation OR gametes / genetic material + combined AW ; 6 grow seeds ; 7 select the ones with best combination of characteristics ;	5	

Question	Answer	Marks	Guidance
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5 (a) Biological processes in plants are affected by environmental factors.

The graphs each show the effect of changing one environmental factor on a biological process.

The labels on the axes for each graph have been replaced with the letters **P**, **Q**, **R** and **S**.



Identify the correct axis label for each of **P**, **Q**, **R** and **S**.

Select from the following possible labels.

- | | | |
|------------------------|-------------------------------------|------------------------------|
| temperature | carbon dioxide concentration | rate of transpiration |
| light intensity | rate of photosynthesis | air humidity |

P

Q

R

S

[4]

(b) Describe the process of transpiration in the leaf of a plant.

.....

.....

.....

..... [3]

[Total: 7]

5(a)	(P) rate of photosynthesis ; (Q) temperature ; (R) rate of transpiration ; (S) air humidity ;	4	
5(b)	1 evaporation ; 2 surface of / cell wall / film of water / layer of water + mesophyll cells ; 3 water vapour ; 4 air spaces / intercellular spaces ; 5 diffusion / high to low concentration / down concentration gradient ; 6 water + through stomata ;	3	

Question	Answer	Marks	Guidance
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5090/22/M/J/22 Q9

9 (a) Compare the ureter and the urethra in terms of their functions in adult humans.

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..... [4]

MS 5090/22/M/J/22 Q9

9(a)	1 both carry + urine / urea ; 2 ureter <u>from</u> kidney ; 3 ureter <u>to</u> bladder ; 4 urethra <u>from</u> bladder 5 urethra <u>out of</u> body ; } (any two from) 6 male / penis / testis + urethra carries + semen / sperm ;	4	
9(b)	1 selectively / partially / semi + permeable ; 2 large surface area ; 3 fluid / dialysate ; 4 counter-current / continuously changed ; 5 urea / toxins / poisons / harmful substances + from blood / to fluid ; 6 water / salts / ions + between blood and fluid ; 7 diffusion / osmosis ; 8 high to low concentration / <u>down</u> concentration gradient ; 9 large <u>molecules</u> / <u>proteins</u> + stay in blood / do not enter fluid ;	6	

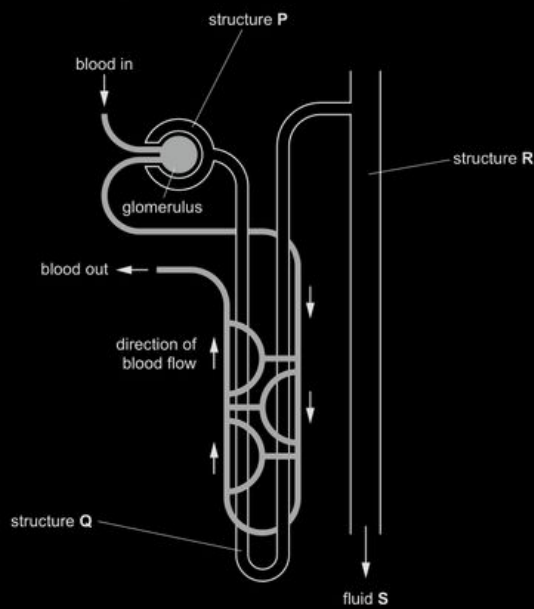


Fig. 4.1

- (a) (i) Identify each of the structures labelled P, Q and R in Fig. 4.1.
- P
- Q
- R [3]
- (ii) Name the organ which contains many nephrons.
- [1]
- (iii) Name the organ that stores fluid S before it is released from the body.
- [1]

- (b) Table 4.1 shows the composition of a sample of fluid S from a healthy person.
- The person that provided the sample of fluid S eats a balanced diet.

Table 4.1

component	concentration/arbitrary units
glucose	0.00
protein	0.00
ions	1.50
urea	2.00

- (i) State why it is important that urea is removed from the blood in fluid S.
- [1]
- (ii) Use your knowledge of the function of a nephron to explain the concentration of each of the following components of fluid S.
- glucose
-
-
-
-
- protein
-
-
- [5]
- (iii) Explain what would happen to the concentration of ions in fluid S if the person drank a large volume of water several hours before the sample was collected.
-
-
-
- [3]

Total: 141

4(a)(i)	(P) Bowman's capsule ; (Q) loop of Henle ; (R) collecting duct ;	3	
4(a)(ii)	kidney ;	1	
4(a)(iii)	(urinary) bladder ;	1	
4(b)(i)	toxic / toxin ;	1	
4(b)(ii)	<i>glucose</i> 1 enters Bowman's capsule / enters nephron / removed from blood AW ; 2 by filtration / filtered; 3 back into blood / reabsorbed + into blood ; 4 AVP ; <i>protein</i> 5 doesn't enter Bowman's capsule / doesn't enter nephron / no filtration / not removed from blood AW ; 6 protein + large AW ; max 5	5	
4(b)(iii)	ion concentration + decrease ; more water / higher volume of water / excess water + in fluid S / urine / excreted; homeostasis / to keep concentration of water in blood constant AW;	3	

5990/22/M/J/24 Q6

6 Flowering plants are sometimes grown in soil to provide cut flowers that can then be displayed.

(a) A gardener watered flowering plants with a fertiliser solution containing dissolved chemicals.

Table 6.1 shows the information printed on the label of the container of fertiliser powder used to prepare this fertiliser solution.

Table 6.1

chemical component	mass/g per 100 g fertiliser powder
nitrates	4.0
phosphates	4.0
sulfur	1.0
magnesium	0.5

- (i) To prepare the fertiliser solution, the gardener dissolved 20 g of the fertiliser powder in 10 dm³ of water.
- Calculate the total mass of sulfur in the fertiliser solution prepared by the gardener.
- Space for working.

..... [2]

- (ii) Explain which chemical component of the fertiliser solution is the most important to plants for:
- making the leaves of the plants green in colour
-
-
- the production of enzymes by cells in the leaves of the plants.
-
- [4]

- (iii) The chemical components of the fertiliser solution are absorbed by the root hair cells of the plants and are then transported to the leaves.
- Complete the words below to name the three main tissues in the pathway taken by the chemical components of the fertiliser solution from the root hair cells to the leaves.
- The first letter of each word has been completed for you.
- C
- ↓
- X
- ↓
- M [3]

- (b) When flowers from the plants are cut and used for display, the leaves are removed before each stem is placed in water.
- A solution containing sucrose is often added to this water.
- Sucrose is **not** needed in fertiliser solution used to water plants growing in soil.
- Explain the benefit of providing sucrose to cut flowers.
-
-
-
-
- [3]

[Total: 12]

6(a)(i)	0.2 ; g ;	2	
6(a)(ii)	magnesium ; <u>chlorophyll</u> (makes leaves green); nitrates ; (enzymes made from) amino acids / proteins ;	4	
6(a)(iii)	(c)ortex ; (x)ylem ; (m)esophyll ;	3	
6(b)	1 no + photosynthesis ; 2 to produce glucose / glucose not produced ; 3 to produce sucrose / sucrose not produced ; 4 sucrose travels in phloem ; 5 for respiration / energy ; 6 flowers / plants + last longer / for days AW ; max 3	3	

(a) Explain how the structure of a leaf is adapted for photosynthesis.

(b) Transpiration is both necessary for plants and a problem for plants.

Discuss this statement.

[Total: 10]

7(a)	1 large surface area + maximum light absorption ; 2 stomata + gaseous exchange / carbon dioxide in / oxygen out ; 3 thin + diffusion of gases / carbon dioxide in / oxygen out ; 4 air spaces / spongy mesophyll + diffusion of gases / gaseous exchange ; 5 chloroplasts / chlorophyll + <u>absorb</u> light ; 6 chloroplasts / chlorophyll + most dense in palisade (mesophyll) ; 7 palisade in upper leaf + receive most light ; 8 xylem (vessels) + bring water (from soil) ; 9 epidermal cells no chloroplasts / transparent + light reaches cells ; max 6	6	
7(b)	necessary (max 2): 1 provides (leaves with) water for photosynthesis ; 2 transports minerals / ions from roots / to leaves ; 3 provides plant cells with support / turgidity / turgor ; 4 cools the leaf / plant ; problem (max 2): 5 too much water is lost (through stomata / leaf) ; 6 (water loss) leads to wilting / death ; 7 plants may close stomata to prevent transpiration, which slows down photosynthesis ; max 4	4	

8 (a) urea;
carbon dioxide;
water;
salts;
toxins/broken-down hormones;
bile salts/pigments; [max 3]

(b) urea/water/salts/toxins/broken down hormones + kidneys;
blood/blood vessels/named vessel/capillaries;
bladder + urine/urination;
ureter + urethra (both correctly spelt);
water/CO₂ + lungs/alveoli;
diffusion + from capillaries (for CO₂);
breathing (out);
water/salts/urea + sweat;
sweat glands;
from blood/capillaries;
sweat ducts/pores;
ref. faeces ONLY in an explanation of how bile salts/pigments are removed; [max 7]

[Total: 10]

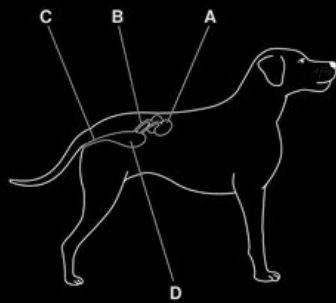


Fig. 1.1

- (a) (i) Name the parts **A**, **B** and **C** in Fig. 1.1.
- A**
- B**
- C** [3]
- (ii) State the function of **D**.
- [1]

- (b) Fig. 1.2 shows the relationship between an alveolus and a blood capillary in the lung of a mammal. The arrows show the path taken by oxygen from inhaled air entering the blood during gas exchange.

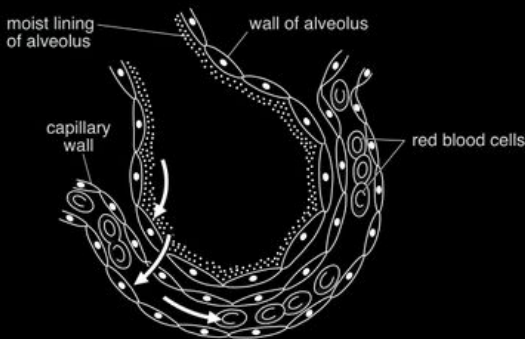


Fig. 1.2

- (i) Draw **two** arrows on Fig. 1.2 to show the path taken by most carbon dioxide leaving the blood.
-
- [2]
- (ii) Explain why this movement of carbon dioxide is considered to be part of excretion.
-
- [3]
- (c) Hormones are also substances that need to be excreted from the body after they have carried out their functions.
Explain how this is done.
-
- [2]

[Total: 11]



- 1

(a)

(i)

A

kidney;

B

ureter;

C

urethra;

(ii)

D

storage AW + urine;

[1]
- (b)

(i)

arrow shown leaving plasma;
arrow from moisture to alveolar air;
One arrow covering the entire journey = 1 mark

[2]
- (ii)

removal + from body / organism;
waste;
of respiration / metabolism;

[3]
- (c)

destroyed / broken down **AW**;
by liver;
products removed / reabsorbed;

[max 2]

[Total: 11]

(a) Explain why phloem is more suitable for this purpose than any other plant tissue.

[3]

[7]

[Total: 10]

7 (a) Explain the differences between *mitosis* and *meiosis*. Your explanation should

- 6 (a) within reach;
contains sucrose / sugar / carbohydrate;
contains amino acids;
in solution / water;
not lignified / is softer **AW**;
R starch / glucose / mineral ions
Ig nutrients

[3]

- (b) impaired growth **AW** / death;
amino acids used to make protein;
less / no new cytoplasm;
less sucrose or carbohydrate;
less glucose;
for use in respiration;
energy used;
for any named purpose;
may introduce viruses / disease;
damage to plant **AW**;
Ig wilting

[max 7]

[Total: 10]

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[Total: 10]



- 7 (a) removal from organism/body;
toxic/poisonous;
waste (products);
from metabolism or described;

[Max 3]

- (b) filters/removes substances from + blood;
using partially permeable membrane **AW**;
ref. dialysis fluid;
urea/nitrogenous products;
salt(s)/ions/small molecules;
toxins/poisons;
ref. relative concentrations **AW**;
excess water/ref. osmoregulation;
large molecules stay in blood;
such as proteins;
ref. diffusion;

[Max 7]

[Total: 10]

7

pac

A amino acids / glucose / sugar

[3].



7 (a) CO₂ / water;

from body **AW**;

waste product / toxic material;

respiration;

metabolic (process);

[max 3]

(b) homeostasis / osmoregulation;

salts / ions / minerals / urea;

water;

from blood;

ref. reabsorption into blood;

in varying quantities / excess;

depending on food/water intake;

and on loss in sweating;

tissue fluid therefore at constant concentration;

water would otherwise enter or leave cells;

osmosis / diffusion;

effect on cell appearance (e.g. swell / burst / shrink);

affect on enzyme action / metabolism;

[max 7]

[Total 10]

5090/21/01/14 02
2 Fig. 2.1 shows the digestion, absorption and assimilation of a chemical (B). It also shows the formation of its waste product, chemical F.

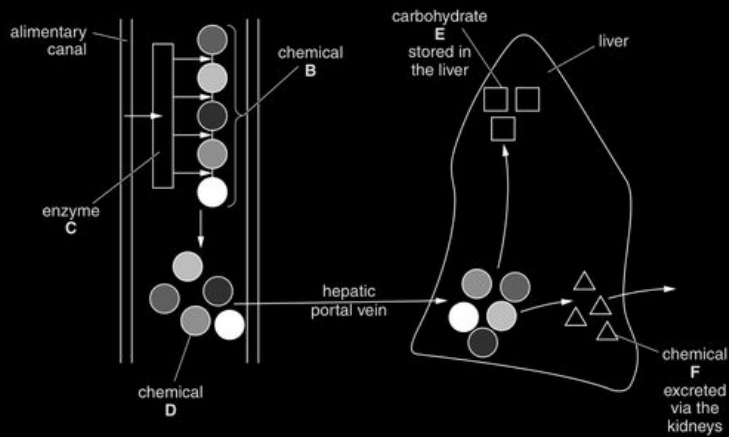


Fig. 2.1

- (a) Name
- chemical B [5]
- enzyme C [5]
- chemical D [5]
- carbohydrate E [5]
- chemical F [5]
- (b) (i) Describe what happens to carbohydrate E before it can be used in metabolic reactions in body cells.
- [2]
- [2]
- (ii) Name the process in which the product of carbohydrate E is used within a body cell.
- process [1]
- (iii) Describe how different types of body cells and tissues make use of this process.
- [4]
- [4]
- [Total: 12]

3 (a) At puberty, hormones are produced that are responsible for the development of secondary

2	(a)	B = protein / (poly)peptide ; C = protease or named* ; D = amino acids* ; E = <u>glycogen</u> ; F = urea ;	[5]	*A ecf for incorrect substrate *A ecf for incorrect substrate R urine
	(b) (i)	broken down / converted / changed ; (role of) glucagon / adrenaline ; to <u>glucose</u> ; made soluble ;	[max. 2]	
	(ii)	<u>respiration</u> ;	[1]	

7 Describe the similarities and differences in structure and function of root hairs and villi.

(a) similarities

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.....

.....

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.....

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.....

..... [4]

(b) differences

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..... [6]

[Total: 10]

7	(a)	<i>structural similarities:</i> long / elongated ; providing large surface area ; <i>functional similarities:</i> absorption / uptake ; active transport / diffusion ; ions / salts / minerals / named ; water ;	[max. 4]	
	(b)	<i>structural differences:</i> cell wall / no cell wall ; (root hair) part of one cell ; (villi) many cells / multicellular ; ref. absence of blood vessels / lacteals / ORA ; AVP ; <i>functional differences:</i> root hairs + absorb from the soil ; villi + absorb from the gut ; villi + absorb amino acids ; villi + absorb glucose ; villi + absorb lipids / glycerol / fatty acids ;	[max. 6]	
			[Total: 10]	

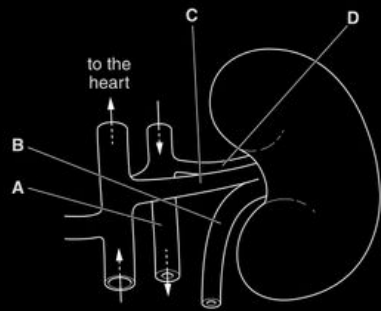


Fig. 1.1

- (a) (i) Name the structure labelled **A** in Fig. 1.1. [1]
- (ii) Name the chamber of the heart through which blood in structure **A** last passed.
..... [1]

- (b) Table 1.1 shows the relative concentrations of various substances in structures **B** and **C**. Complete the table to show the possible concentrations of these substances in structure **D**.

Table 1.1

substance	relative concentration in structure		
	B	C	D
amino acids	0.00	0.05	
glucose	0.00	0.10	
mineral ions	1.50	0.72	
proteins	0.00	8.00	
urea	2.00	0.03	

[5]

- (c) Explain how the relative concentrations of glucose might change in structures **B**, **C** and **D** in a person with diabetes.
.....
.....
.....
.....
.....
..... [3]

[Total: 10]

1	(a) (i)	(dorsal) <u>aorta</u> ;	[1]	
	(ii)	<u>left ventricle</u> ;	[1]	
	(b)	(amino acids) 0.05 ; (glucose) 0.10 to 0.15 ; (mineral ions) 0.72 to 2.22 ; (proteins) 8.00 ; (urea) 0.03 to 2.03 ;	[5]	
	(c)	B would contain some / more / high (glucose) / C would contain more / high (glucose) / D would contain more / high (glucose) ; lack of Insulin ; glucose would not be converted into glycogen ; kidney unable to/doesn't reabsorb all glucose ;	[max. 3]	
			[Total: 10]	



- Suggest an explanation for this observation.

- Complete the equation below for photosynthesis using either words or symbols.



[1]

-
- The diagram illustrates the structure of a leaf. On the left, a whole leaf is shown with a central vein and lateral veins. A small section of the leaf is magnified on the right, showing a single mesophyll cell. The cell is roughly rectangular with a thick cell wall. Inside the cell, there is a large central vacuole (labeled P) that occupies most of the space. A nucleus (labeled Q) is located near the center. A chloroplast (labeled R) is situated near the cell wall, containing internal stacks of thylakoids.

Fig. 5.2

- P**

9

R

[3]

[Total: 10]

5 (a)	<i>wilting:</i> root hairs removed / roots damaged ; stem / xylem damaged ; less water absorbed ; plant loses water / transpires ; cells lose water / cells become flaccid ; <i>recovery:</i> root (hairs) regrow ; (more) water absorbed ; cells gain water again / become turgid ;	[max. 4]	R no water absorbed allow max.3 marks for <i>wilting</i> answers
(b)	carbon dioxide + water → oxygen ;	[1]	accept words or chemical formulae or a correctly balanced symbol equation I light / chlorophyll
(c) (i)	P = (lower) epidermal cell / epidermis / epidermal layer ; Q = guard cell ; R = stoma / stomata ;	[3]	
(ii)	fewer / no stomata / guard cells ; reduce / stop water loss / transpiration ; (appears) shiny / bright / ref. to cuticle ;	[max. 2]	
		[Total: 10]	

(a)

[3]

(b)

[7]

[Total: 10]

9 (a)	removal from organism/body ; toxic/poisonous ; waste (products) ; from metabolism ;	[max. 3]	
(b)	1 <u>blood</u> from patient to machine/returned to patient ; 2 using <u>partially permeable</u> membrane/tubes AW ; 3 ref. dialysis fluid ; 4 ref. relative concentrations AW ; 5 filters/removes substances from + blood /unwanted out /wanted in ; 6 by <u>diffusion</u> ; 7 urea / nitrogenous products / toxins / poisons ; 8 salt(s) / ions / small molecules ; 9 excess water / ref. osmoregulation ; 10 large molecules / proteins stay in blood ; 11 practical aspect e.g. bubble trap / repetition / frequency / duration of procedure / dialysis fluid frequently changed / temperature ref / counter flow ;	[max. 7]	
		[Total: 10]	

7 (a) (i) Explain how wilting occurs in a plant.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[4]

(ii) Explain why mature plants do not wilt as much as newly-transplanted seedlings, even when both are exposed to similar conditions.

.....

.....

.....

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.....

.....

.....

.....

.....[3]

(b) Suggest an explanation for what would happen to a plant if it is accidentally ‘watered’ with a concentrated salt solution.

.....

.....

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.....[3]

[Total: 10]

7	(a) (i)	water lost / plant dehydrated ; by transpiration / evaporation / through stomata ; faster than it is gained ; cells / tissues lose turgor / become flaccid / plasmolysed AW ; no longer support plant ; any condition leading to wilting ;	[max 4]	A lack of water, low humidity, high light intensity, high temperature, wind
	(ii)	(mature plant) root system undamaged ORA ; ref. root hairs ; water uptake sufficient for plant ORA ; plant cells / tissues turgid ORA ; xylem supports older plants / ref. lignin ;	[max 3]	I ref. more / less water uptake
	(b)	plant / leaves will wilt ; water lost ; from cells / tissues ; by osmosis / diffusion ; lower water potential in soil water / ref. correct water potential gradient AW ;	[max 3]	
			[Total: 10]	

2 Fig. 2.1 shows a section through a human kidney and its associated structures.

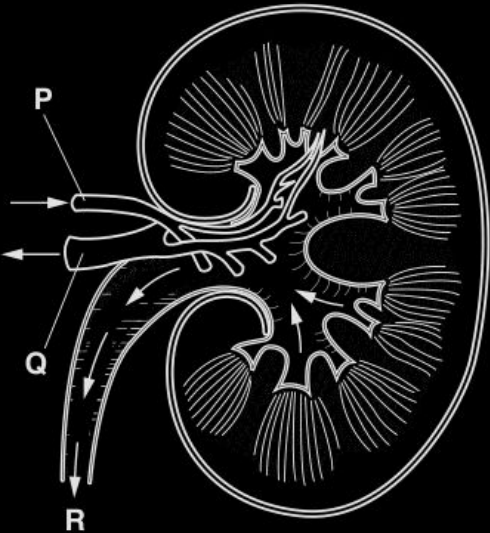


Fig. 2.1

(a) (i) Name the blood vessels **P** and **Q** in Fig. 2.1.

P

Q

[2]

(ii) Name the fluid **R** in Fig. 2.1 and describe the route taken by this fluid from leaving the kidney to being removed from the body.

name of fluid

route taken by fluid

.....

.....

.....

[4]

(b) A person with kidney damage requires the process of dialysis to take place several times

2(a)(i)	P renal artery ; Q renal vein ;	2	
2(a)(ii)	urine ; ureter ; bladder ; urethra ;	4	R urea
2(b)(i)	reference to artery + vein ;	1	
2(b)(ii)	lower pressure ; less / no urea ; reference to increase / decrease + concentration of ions AW ;	1	
2(c)	(artificial membrane) partially permeable AW ; small molecules or named + pass through (membrane) ; correct reference to diffusion / osmosis ; proteins do not pass through (membrane) ; (dialysis fluid) no / low urea OR removes urea ; reference to concentration gradient ; same glucose concentration + as normal blood ; same ion AW concentration + as normal blood ;	5	

(b) Explain why carbon dioxide does not normally pass out of the leaves of a plant during the day.

[Total: 10]

9 (a) Describe and explain how a nitrate ion in the soil becomes part of a stored molecule in the

8(a)	digested / broken down /hydrolysis ; into amino acids ; by protease or named enzyme ; absorbed in small intestine, duodenum / ileum ; in villi / by capillaries / blood ; hepatic portal vein ; to liver ; deamination or described ; urea + in renal artery ; urea + taken to kidney ; blood filtered / urea leaves blood ; forms urine ; bladder / urethra ;	8	
------	--	---	--

8(b)	photosynthesis ; uses up CO ₂ ; faster than is made during respiration ;	2	
------	---	---	--

Question	Answer	Marks	Guidance
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7 The kidneys are organs of excretion.

(a) Define the term *excretion*.

.....

.....

.....

.....

..... [3]

7(a)	1 removal AW ; 2 toxins / toxic ; 3 waste products AW / urea / carbon dioxide ;	3	3 R egestion
7(b)(i)	1 multiple tubes / multiple sheets / coiled / folded / long ; 2 large AW + surface area ; 3 selectively / partially / semi + permeable ; 4 diffusion / osmosis ; 5 small / waste products AW / urea / ions / water / toxins + do pass AW ; 6 large / protein / blood cells + do not pass AW ;	5	

7(b)(ii)	1 maintain gradient AW ; 2 prevent build up / ensure removal + waste products AW / urea / toxins ; 3 continued / more / maximum + diffusion / movement / exchange ;	2	
----------	---	---	--

Question	Answer	Marks	Guidance
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3 A man is cooking in a hot kitchen and sweat is being released by sweat glands in his skin.

(a) (i) Explain the term homeostasis with reference to sweating.

.....

.....

.....

..... [2]

(ii) Explain how receptors and blood vessels in the man's skin are also involved in homeostasis when he is in the hot kitchen.

.....

.....

.....

.....

.....

..... [4]

(b) Sweat contains very small quantities of the waste product urea.

State the name of the organ where most urea is made in the body.

..... [1]

(c) The table shows the concentrations of some chemicals found in the sweat, urine and blood plasma of a healthy human.

chemical	concentration of chemical /mmol per dm ³		
	sweat	urine	blood plasma
urea	22	393	6
sodium	66	110	141
chloride	59	103	99

(i) Use the data in the table to compare the chemical composition of urine and blood plasma.

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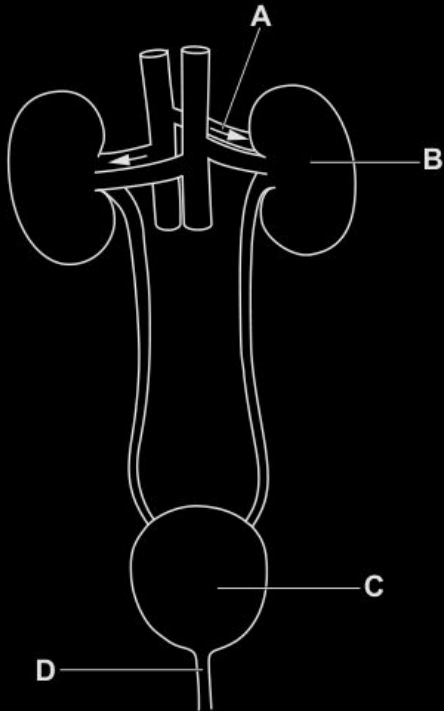
..... [4]

3(a)(i)	<i>any two from:</i> (homeostasis is) the maintenance of a <u>constant</u> ; <u>internal environment</u> / temperature ; return to norm / return to set-point / negative feedback ; <u>evaporation</u> of sweat removes heat / thermal energy / cools / decreases temperature ;	2	
3(a)(ii)	<i>any four from:</i> receptors + detection of temperature ; <u>impulse</u> to brain / CNS ; <u>vasodilation</u> ; arterioles / blood vessels widen ; increase in blood + near skin surface ; greater + loss of heat / thermal energy ; by radiation ;	4	
3(b)	<u>liver</u> ;	1	

3(c)(i)	<i>any four from:</i> less urea in blood plasma ; more sodium in blood plasma ; less chloride in blood plasma ; comparative data quote with units ; data manipulation, e.g. 50 × higher urea in urine than blood plasma ;	4	
3(c)(ii)	high(er) / over 22 (in patient) ; reduced filtration / reduced removal of urea (from kidney) ; reference to blood <u>plasma</u> ;	3	

Question	Answer	Marks	Guidance
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1 The diagram shows parts of the human body involved in excretion.



(a) Name the liquid that passes out of the human body through tube **D** and the waste product it contains.

liquid

waste product

[2]

(b) Describe how parts **A**, **B** and **C** are involved in removing the waste product that passes out of tube **D**.

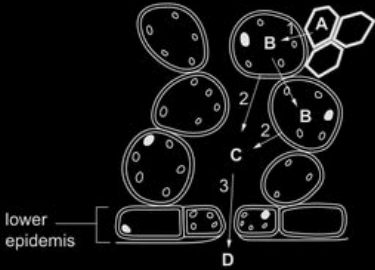
.....
.....
.....
.....
..... [3]

[Total: 5]

2 Grapes are fruits that grow on a plant called a grape vine.

1(a)	urine ;	1	
	urea ;	1	A ammonia / excess water / excess minerals
1(b)	renal artery / A + brings urea / waste product to B / kidney AW ;	1	
	kidney / B + <u>filters</u> blood to remove urea / waste products ;	1	
	bladder / C + stores / keeps / collects <u>urine</u> ;	1	
Question	Answer	Marks	Guidance

6 The diagram shows some cells from the lower part of a transpiring leaf.



- (a) (i) Identify cell type B.
- B [1]
- (ii) Water travels from A to D. The arrows on the diagram represent the movement of water molecules during transpiration. With reference to these arrows, name and define each process taking place.
- arrow 1
.....
.....
- arrow 2
.....
.....
- arrow 3
.....
..... [6]
- (b) Leaves from different plant species transpire at different rates in the same environmental conditions.
Suggest and explain structural differences in the lower surface of these leaves which might produce differences in the transpiration rate.
-
.....
.....
.....
..... [3]
- [Total: 10]

6(a)(i)	<u>spongy mesophyll</u> ;	1	
6(a)(ii)	<i>arrow 1</i> osmosis ; is the movement (of water molecules) from an area of higher water potential to lower water potential across a partially permeable membrane AW ; <i>arrow 2</i> evaporation ; is the change from liquid (water) to (water) vapour / gas ; <i>arrow 3</i> diffusion ; is the (net random) movement (of molecules) from a high to lower concentration / down a concentration gradient ;	6	A correct definition if linked to correct process but linked to the incorrect arrow.

6(b)	the surface area of the leaf ; the greater the surface area of the leaf, the greater the surface for evaporation ; the thickness of the cuticle ; the thicker the cuticle, the greater the barrier for water molecules ; the number / density of stomata ; as density of stomata increase, more places for water to escape from ; size of stomata ; as size of stomata increases, so does area from which water can be lost ; sunken stomata / ridged epidermis / curled up leaves ; sunken stomata / ridged epidermis / curled up leaves increase humidity ; hairs / projections ; hairs / projections increase humidity ;	3	Maximum two marks for structural features only with no explanation.
------	---	---	---

9

(a) Explain the advantages and disadvantages of the use of insecticides in agriculture.

8(a)	<i>day</i> 1 light / day + photosynthesis ; 2 faster than respiration / respiration uses up oxygen ; 3 day + CO₂ used / in + O₂ produced / out ; <i>night</i> 4 no light / night + no photosynthesis ; 5 respiration ; 6 night / evening + O₂ used / in + CO₂ produced / out ; 7 stomata + open in day / closed at night ; 8 more water vapour lost during day ; 9 compensation point(s) AW ;	5	
8(b)	1 xylem ; 2 from xylem into mesophyll ; 3 diffusion / osmosis ; 4 down water potential gradient AW ; 5 layer of water + mesophyll + cell wall / outside cell ; 6 evaporation / water vapour ; 7 air / <u>intercellular</u> + spaces ; 8 exits leaf through stomata / stoma ;	5	

7 (a) A root hair cell in a plant is respiring aerobically.

(i) Name the structures within the cell that carry out aerobic respiration.

..... [1]

(ii) Write the balanced symbol equation for the aerobic respiration of glucose, $C_6H_{12}O_6$.

..... $\longrightarrow$ [2]

(b) The plant experiences changing weather conditions including:

high temperature heavy rain high winds low humidity

State the weather condition that may lead to a root hair cell respiring anaerobically and explain your answer.

weather condition

explanation

..... [2]

[Total: 5]

8 (a) The number of individuals in a population can vary with time.

7(a)(i)	mitochondria / mitochondrion ;	1	
7(a)(ii)	$C_6H_{12}O_6 + 6O_2 ;$ $6H_2O + 6CO_2 ;$	2	A correct word equation for max 1 mark Ig any reference to energy or ATP
7(b)	heavy rain ; soil waterlogged / AW + no air spaces / oxygen available ;	2	

3 Fig. 3.1 shows diagrams that represent the blood flow into and out of three major organs through blood vessels P to U.

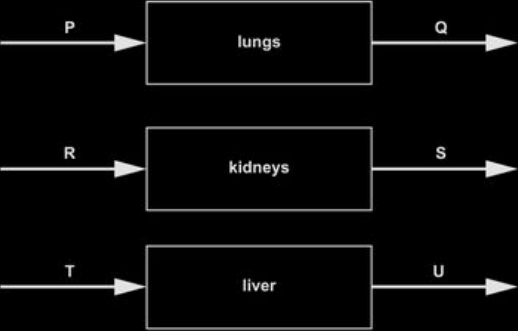


Fig. 3.1

(a) Identify each of the following blood vessels.

- Q [3]
R
U

(b) Compare the structures of blood vessels Q and R.

.....
.....
.....
.....
..... [3]

(c) Explain each of the following.

(i) Blood in vessel S has a lower concentration of urea than blood in vessel R.

.....
.....
.....
.....
..... [3]

(ii) Blood in vessel U sometimes has a higher concentration of glucose than blood in vessel T.

.....
.....
.....
.....
..... [3]

[Total: 12]

4 Fig. 4.1 shows a Siamese fighting fish that lives in fresh water.

3(a)	(Q) <u>pulmonary vein</u> ; (R) <u>renal artery</u> ; (U) <u>hepatic vein</u> ;			3																
3(b)	<table border="1"><tr><td>Q – (pulmonary) vein</td><td></td><td>R – (renal) artery</td></tr><tr><td>thin + wall</td><td>+</td><td>thick + wall ;</td></tr><tr><td>thin muscle / thin elastic</td><td>+</td><td>thick muscle / thick elastic ;</td></tr><tr><td>wide + lumen AW</td><td>+</td><td>narrow + lumen AW ;</td></tr><tr><td>valves</td><td>+</td><td>no valves ;</td></tr></table>			Q – (pulmonary) vein		R – (renal) artery	thin + wall	+	thick + wall ;	thin muscle / thin elastic	+	thick muscle / thick elastic ;	wide + lumen AW	+	narrow + lumen AW ;	valves	+	no valves ;	3	/
Q – (pulmonary) vein		R – (renal) artery																		
thin + wall	+	thick + wall ;																		
thin muscle / thin elastic	+	thick muscle / thick elastic ;																		
wide + lumen AW	+	narrow + lumen AW ;																		
valves	+	no valves ;																		
3(c)(i)	1 urea + into kidney OR kidney removes + urea ; 2 filtration / filtering ; 3 nephron / Bowman's capsule / glomerulus ; 4 urea + excreted / in urine ;			3																

3(c)(ii)	1 (blood) glucose decreased / sugar low / body needs more sugar ; 2 glycogen ; 3 breakdown / hydrolysis + to glucose ; 4 glucose + passed out of liver / goes into U / hepatic vein / blood ; 5 (stimulated by) glucagon ;		
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Question	Answer	Marks	Guidance
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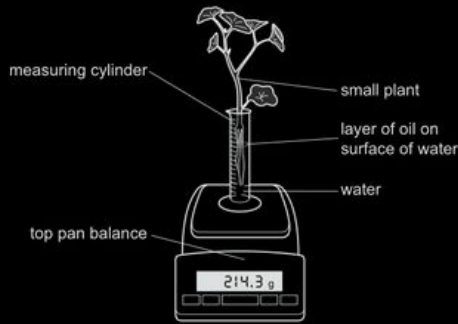
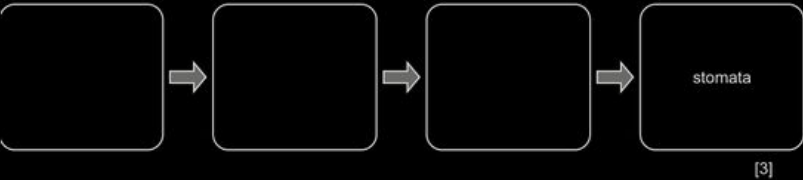


Fig. 1.1

- (a) (i) Name the process by which a plant loses water from cells to the atmosphere.
..... [1]
- (ii) Suggest the purpose of the layer of oil.
.....
..... [1]
- (iii) Describe **one** way in which a student could use this apparatus to obtain a value for the amount of water the plant loses in 24 hours.
.....
.....
..... [2]

- (b) Water is lost mainly through the plant's stomata.
Name **three** types of cell that the water passes through before it reaches the stomata in the leaves.
Write the names of the types of cell in the correct order in the boxes.



- (c) The experiment shown in Fig. 1.1 was repeated many times to investigate the effect of humidity on the rate of water loss.
All environmental conditions were kept constant except for humidity which was gradually increased.
Fig. 1.2 shows the axes for a graph.

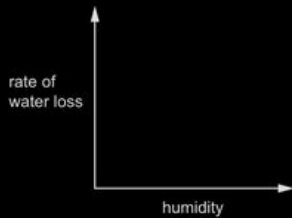


Fig. 1.2

- (i) Sketch a graph on these axes of the expected results for this investigation. [1]
- (ii) Explain what your graph shows about the effect of humidity on water loss.
.....
.....
.....
..... [3]

[Total: 11]

1(a)(i)	transpiration ;	1	
1(a)(ii)	prevent water loss / evaporation of water;	1	
1(a)(iii)	measure mass / volume at start + at end / after 24 hours ; subtract end value from start value / calculate difference ;	2	
1(b)	any three in correct sequence from: root hair (cell) ; (root) cortex (cell) ; xylem (vessel element) ; palisade / spongy / mesophyll (cell) ;	3	
1(c)(i)	line that is at its highest at a low humidity and at its lowest at a high humidity ;	1	
1(c)(ii)	the lower the humidity, the higher the rate of water loss AW ; water is lost by diffusion ; when humidity is low, the concentration of water outside will be lower than inside the leaf / atmosphere would not be saturated with water ;	3	

4 Living organisms excrete carbon dioxide.

(a) (i) Describe the meaning of the term excretion.

.....

.....

..... [2]

(ii) Name the process which produces carbon dioxide in living organisms and state how carbon dioxide is excreted by humans.

.....

.....

..... [2]

(b) The volume of carbon dioxide excreted varies over a 24-hour period and the factors that affect this are different for animals and plants.

Suggest and explain reasons why the volume of carbon dioxide excreted varies for both animals and plants.

animals

.....

.....

plants

.....

..... [4]

(c) Explain the effect of an increasing concentration of carbon dioxide in the atmosphere.

.....

.....

.....

..... [2]

[Total: 10]

5 The castor oil plant is a flowering plant which produces seeds.

4(a)(i)	any two from: removal + waste products / AW ; removal + toxic materials ; removal + substances made by the body / metabolism ;	2	
4(a)(ii)	<u>respiration</u> ; breathed out / exhaled / removed at the lungs / alveoli ;	2	

4(b)	<i>animals</i> activities vary / example given ; so more respiration at some times ; <i>plants</i> respiration + constant / ongoing ; daylight + photosynthesis AW ; carbon dioxide used for photosynthesis ;	4	max two marks for plants
4(c)	any two from: acts as a greenhouse gas ; traps thermal / heat energy / global warming ; climate change / specific effect on land or organisms ;	2	

Question	Answer	Marks	Guidance
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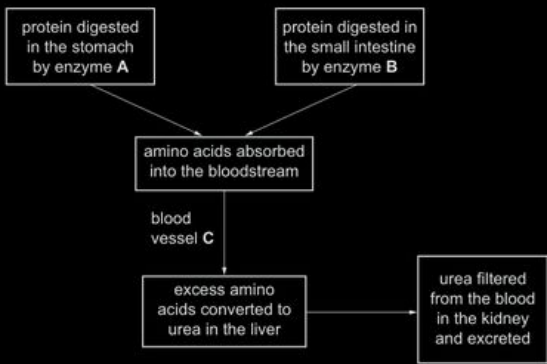


Fig. 3.1

Use Fig. 3.1 to answer these questions.

(i) State the name of enzyme A and enzyme B.

enzyme A

enzyme B

[2]

(ii) State the name of blood vessel C.

..... [1]

(iii) Name the capillary structures in the kidney that filter urea from the blood.

..... [1]

(b) Two groups of students, X and Y, investigate the effect of different diets on the concentration of urea in their urine.

For one day, the two groups eat different foods:

- group X eat bread, jam, orange juice, lettuce, fruit
- group Y eat nuts, milk, bread, cheese, meat.

Early the next morning, they collect and analyse their urine to measure the concentration of urea.

Table 3.1 shows their results.

Table 3.1

	concentration of urea /g per 100 cm ³ of urine						mean
	individual students						
group X	1.5,	1.7,	2.3,	1.6,	2.1,	1.4	1.8
group Y	1.8,	1.9,	2.3,	2.5,	1.9,	2.4	

(i) Calculate the mean concentration of urea for the students in group Y.

Give your answer to **two** significant figures.

Space for working.

mean concentration of urea =g per 100 cm³ of urine [2]

(ii) Explain the difference in the mean concentration of urea for the two groups of students.

.....
.....
.....
.....
.....
.....
..... [3]

(iii) One of the students said: 'To make a fair comparison, we should have made sure that the students in each group drank the same volume of water'.

Explain why this student is correct.

.....
.....
.....
.....
.....
..... [2]

3(a)(i)	pepsin ; trypsin ;	2	
3(a)(ii)	hepatic portal vein ;	1	
3(a)(iii)	glomerulus ;	1	
3(b)(i)	2.1 ;	2	A 12.8 / 6 OR 2.133 = 1 mark
3(b)(ii)	group Y ate more protein / ate protein rich foods ; so / in group Y, more amino acids are deaminated to produce urea ; so / in group Y, more urea is filtered / passes into the urine in the kidney ;	3	A ora for group X
3(b)(iii)	idea that the volume of water drunk, will affect the volume of water used to form urine ; this / the volume of water drunk, will affect the concentration of urea in the urine ;	2	

Question	Answer	Marks	Guidance
----------	--------	-------	----------

[4]

5 (a) Fig. 5.1 shows a model of the human circulatory system.

4(a)	magnesium ; chloroplasts ; palisade (mesophyll) ; sucrose ; cellulose ;	5	
------	---	---	--

4(b)	max four from: mistletoe / it has less chlorophyll ; so can perform less photosynthesis ; therefore, it makes less sugars ; so the mistletoe obtains sugars / sucrose from the tree ; phloem contains sugars / sucrose ;	4	
------	--	---	--

Question	Answer	Marks	Guidance
----------	--------	-------	----------

(b) A scientist measured the mass of water taken up and lost by a plant over 21 hours. The plant was kept at a constant temperature of 20 °C on a windowsill in a laboratory. Readings were taken every 90 minutes.

At 0 hours it was sunrise and at 15 hours it was sunset.

The results are shown in Fig. 2.2.

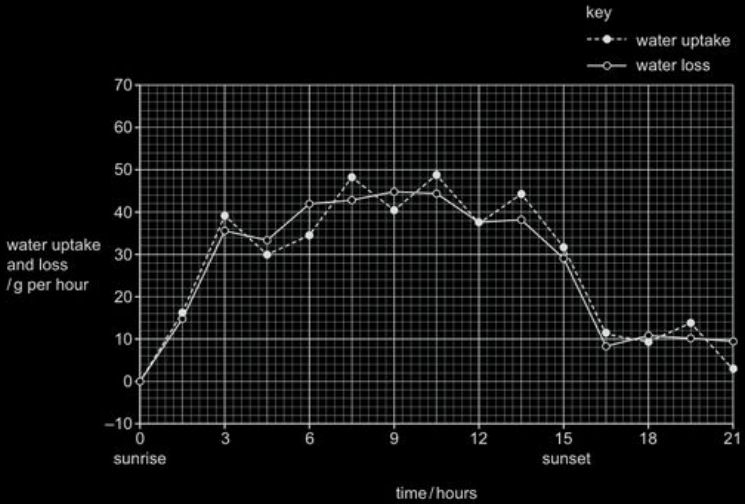


Fig. 2.2

(i) Plants lose water by evaporation from their leaves.

State the name of this process.

..... [1]

(ii) Explain the shape of the curve for **water loss** from the plant over the 21 hours of the experiment.

.....
.....
.....
.....
..... [2]

(iii) Explain why the shape of the curve for water loss is very similar to the shape of the curve for water uptake.

.....
.....
.....
.....
..... [3]

(iv) The scientist repeated the experiment. All conditions were kept the same except the temperature of the laboratory, which was 10 °C lower.

On Fig. 2.2, draw a new curve to show the expected results for **water loss** at the lower temperature. [1]

[Total: 12]

2(a)(i)	8 ;;	2	A 200 ÷ 2500 or 0.08 = 1 mark
2(a)(ii)	max three from: 1 homeostasis ; 2 maintaining constant body temperature ; 3 evaporation of sweat to remove heat (energy) ; 4 increased activity / increase in temperature means more sweat / ora ; 5 if volume lost in faeces / lungs / urine changes, then the percentage lost in sweat will change ;	3	
2(b)(i)	transpiration ;	1	

2(b)(ii)	max two from: increased water loss + during light / day / 0–15 hours / around sunrise + stomata open ; decreased water loss + during dark / night / 15–21 hours / around sunset + stomata close ; greater light intensity + faster evaporation / more diffusion / ora ;	2	
2(b)(iii)	max three from: 1 water evaporates / transpires from leaves ; 2 this reduces water potential of leaves ; 3 water moves up / through xylem vessels ; 4 transpiration pull / transpiration stream ; 5 forces of attraction between water ; 6 prevents loss of turgidity / prevents wilting ;	3	
2(b)(iv)	same shape of curve but below the original ;	1	

Question	Answer	Marks	Guidance
----------	--------	-------	----------

2 By which process does water vapour pass out of a leaf?

- A active transport
- B diffusion
- C osmosis
- D translocation

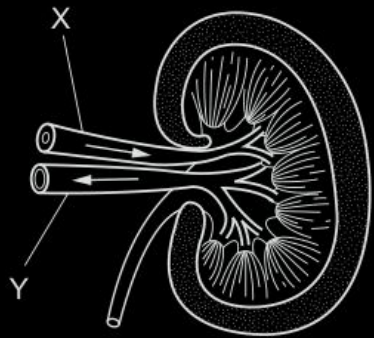
2	B	1
---	---	---

- 11** Which feature of root hairs suggests that they take up ions from the soil by active transport?
- A** Their cell membranes are partially permeable.
 - B** They have a large surface area.
 - C** They have a lower water potential than the soil.
 - D** They take up ions more slowly in low oxygen concentrations.

12 The photomicrograph shows part of a section through a root.

11	D	1
----	---	---

19 The diagram shows a kidney and blood vessels associated with it.



Which statements are correct?

- 1 Vessel X contains more urea than vessel Y.
- 2 Vessel Y contains urine.
- 3 Vessel X contains more oxygen than vessel Y.

A 1, 2 and 3 B 1 and 3 only C 1 only D 2 and 3 only

20 Which of these statements describes control by **negative** feedback?

MS 5090/11/M/J/20 Q19

19	B	1
----	---	---

5090/12/M/J/20 Q2

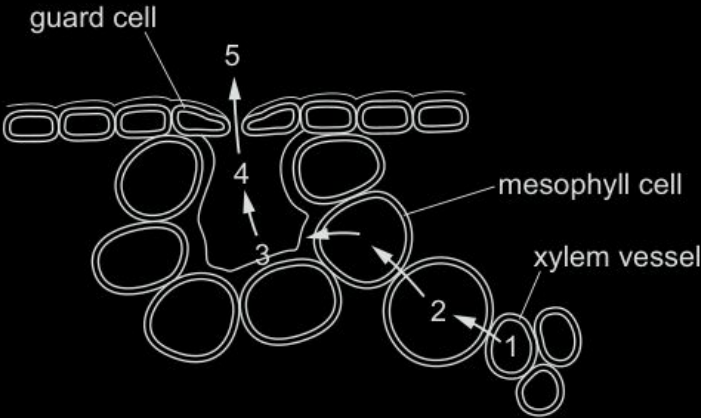
2 Which movement of a substance in plants is an energy-consuming process?

- A absorption of carbon dioxide by a palisade cell
- B absorption of oxygen by a mesophyll cell
- C nitrate uptake by root hair cells
- D transport of water up through the xylem

MS 5090/12/M/J/20 Q2

2	C	1
---	---	---

3 The diagram shows the movement of water through part of a leaf.



Which processes are involved in the movement of water at these stages?

	1–2	3–4	4–5
A	diffusion	evaporation	osmosis
B	evaporation	diffusion	osmosis
C	osmosis	diffusion	evaporation
D	osmosis	evaporation	diffusion

4 Which substance is an enzyme?

3	D	1
---	---	---

5 What is the effect on plants of a lack of magnesium ions?

- A Chlorophyll production is reduced.
- B Leaves turn dark green.
- C Starch production increases.
- D Young leaves wilt.

6 The diagram shows a green plant photosynthesising under a glass jar. This was used as a

5	A	1
---	---	---

11 Water and ions can reach the xylem of a plant root through cell walls, without passing through a cell membrane.

How do these substances move through the cell walls?

	water	ions
A	diffusion	diffusion
B	diffusion	osmosis
C	osmosis	diffusion
D	osmosis	osmosis

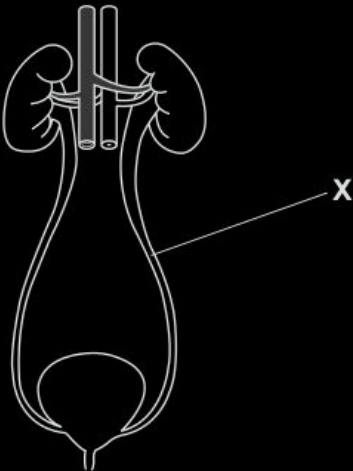
12 The photomicrograph shows part of a section through a root.

MS 5090/12/M/J/20 Q11

11	A	1
----	---	---

5090/12/M/J/20 Q19

19 The diagram shows the human kidneys and their associated structures.



What is the structure labelled X?

- A renal artery
- B renal vein
- C ureter
- D urethra

MS 5090/12/M/J/20 Q19

19	C	1
----	---	---

6 How do carbon dioxide and water enter a leaf?

	carbon dioxide	water
A	diffusion	active transport
B	diffusion	transpiration pull
C	osmosis	active transport
D	osmosis	transpiration pull

MS 5090/11/O/N/20 Q6

6	B	1
---	---	---

5090/11/O/N/20 Q9

9 Blood flows away from the small intestine in the hepatic portal vein.

Where does the blood go next, and what may happen there?

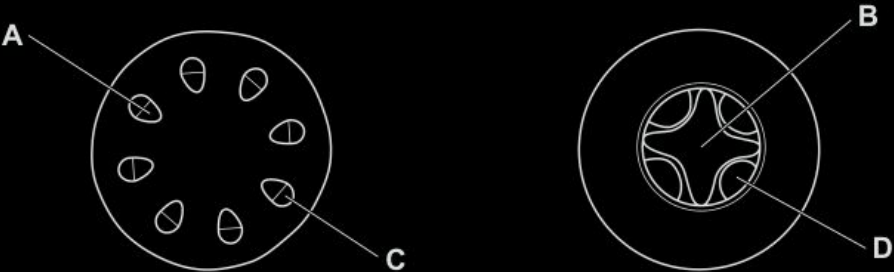
- A the kidneys, delivering glucose for respiration
- B the kidneys, for removal of urea
- C the liver, where alcohol is broken down
- D the liver, where bile is added to the blood

MS 5090/11/O/N/20 Q9

9	C	1
---	---	---

5090/11/O/N/20 Q11

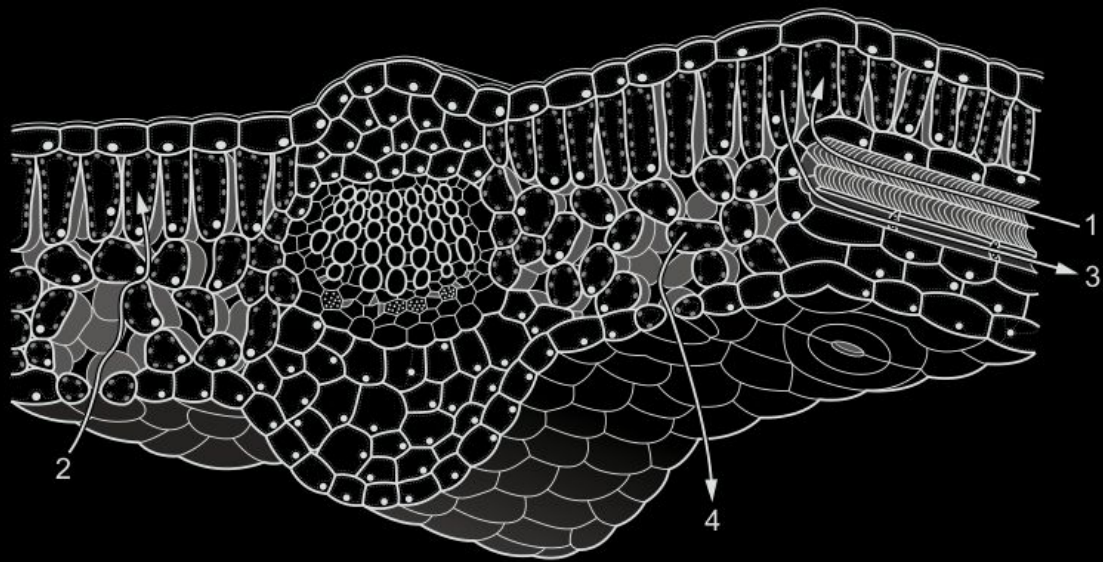
11 Which diagram shows the position of phloem tissue in a herbaceous, dicotyledonous root?



MS 5090/11/O/N/20 Q11

11	D	1
----	---	---

12 The diagram represents part of a leaf.



Which arrows show translocation and transpiration?

	translocation	transpiration
A	1	2
B	2	3
C	3	4
D	4	1

13 In the process of blood clotting, the1..... help to convert2..... to3..... which

MS 5090/11/O/N/20 Q12

12	C	1
----	---	---

5090/12/O/N/20 Q5

5 Which part of a leaf absorbs and uses carbon dioxide from the air?

- A cuticle
- B mesophyll
- C phloem
- D xylem

MS 5090/12/O/N/20 Q5

5	B	1
---	---	---

6 How do carbon dioxide and water enter a leaf?

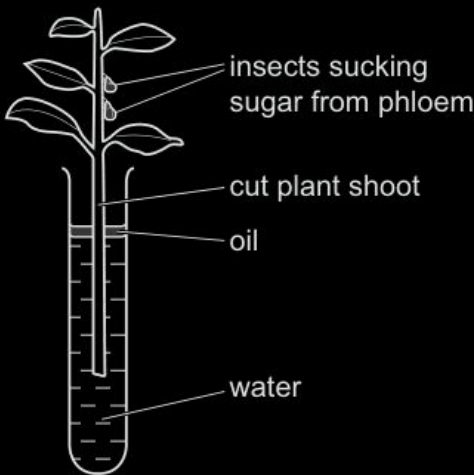
	carbon dioxide	water
A	diffusion	active transport
B	diffusion	transpiration pull
C	osmosis	active transport
D	osmosis	transpiration pull

MS 5090/12/O/N/20 Q6

6	B	1
---	---	---

5090/12/O/N/20 Q12

12 The diagram shows a cut plant shoot in a container of water.



What will stop the movement of water up this stem?

- A a fall in the humidity of the air
- B a rise in the air temperature
- C a blockage in the xylem
- D an insect sucking sugar from the phloem

MS 5090/12/O/N/20 Q12

12	C	1
----	---	---

5090/11/M/J/21 Q11

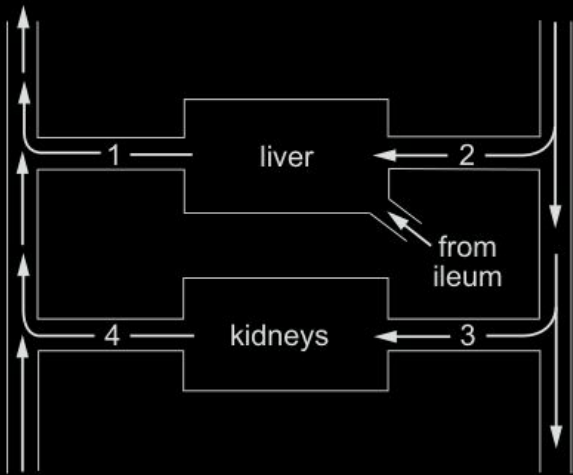
11 Which feature of a root hair cell helps it take up ions from water in the soil?

- A The cell has a partially permeable cell wall.
- B The cell is able to provide a source of energy.
- C The cell vacuole has a high ion concentration.
- D The cell vacuole has a low water potential.

MS 5090/11/M/J/21 Q11

11	B	1
----	---	---

15 The diagram represents the blood supply to the liver and to the kidneys.



Which vessels contain blood with the highest and lowest concentrations of urea?

	highest	lowest
A	1	2
B	1	4
C	3	2
D	3	4

15	B	1
----	---	---

1 Which row is correct?

	cell	tissue	organ	organ system
A	chloroplast	mesophyll	liver	digestive
B	sap vacuole	red blood cell	blood	nervous
C	red blood cell	sap vacuole	stomach	liver
D	neurone	vascular bundle	leaf	excretory

1	D	1
---	---	---

7 A lack of certain mineral ions affects the growth of young plants.

Which effects are caused by a lack of magnesium ions and a lack of nitrate ions?

	lack of magnesium ions	lack of nitrate ions
A	the leaves become dark green	overall growth is stunted
B	the leaves go yellow in patches	the plant is unable to form proteins
C	the plant is unable to form enzymes	the plant is unable to form starch
D	the plant is unable to form root hairs	the leaves become bright green

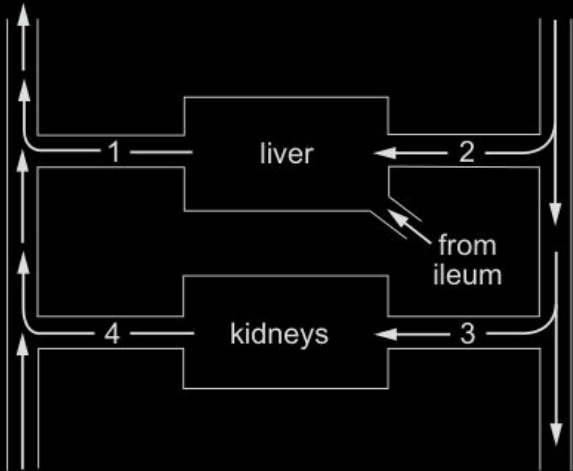
7	B	1
---	---	---

11 How does water enter root hair cells?

- A from high to low water potential by active transport
- B from high to low water potential by osmosis
- C from low to high water potential by active transport
- D from low to high water potential by osmosis

11	B	1
----	---	---

15 The diagram represents the blood supply to the liver and to the kidneys.



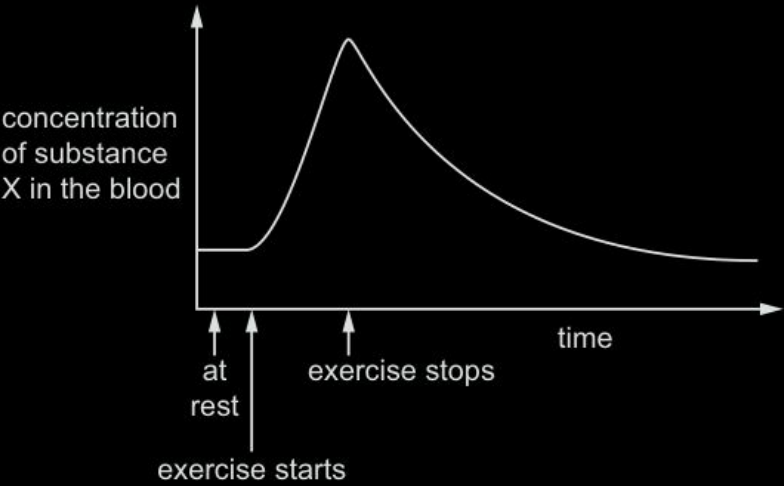
Which vessels contain blood with the highest and lowest concentrations of urea?

	highest	lowest
A	1	2
B	1	4
C	3	2
D	3	4

16 Which statement is true for aerobic respiration?

15	B	1
----	---	---

17 The graph shows the concentration of substance X in a person's blood, before, during and after exercise.



What is substance X?

- A alcohol
- B glycogen
- C lactic acid
- D urea

MS 5090/12/M/J/21 Q17

17	C	1
----	---	---

5090/11/O/N/21 Q2

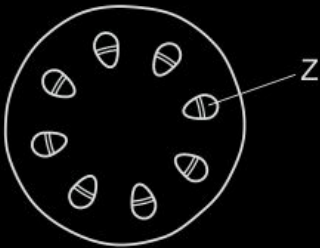
2 Which process occurs by diffusion alone?

- A movement of carbon dioxide into the palisade cells in a leaf
- B uptake of magnesium ions by root hair cells
- C movement of food from the stomach to the duodenum
- D transfer of glucose molecules from the ileum into the blood

3 An experiment to investigate osmosis is set up as shown.

2	A	1
---	---	---

11 The diagram shows a section through a plant stem.



Which process is responsible for moving substances in region Z?

- A diffusion
- B osmosis
- C translocation
- D transpiration

12 Which statement describes why a leaf wilts?

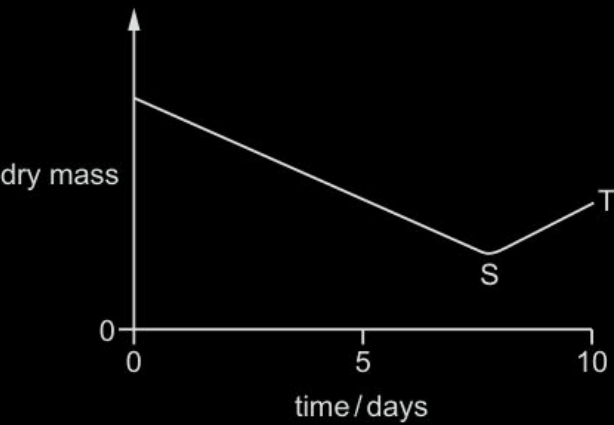
11	C	1
----	---	---

12 Which statement describes why a leaf wilts?

- A Less water is absorbed by the root hairs when the stomata are closed.
- B Less water is absorbed by the root hairs than is lost through the stomata.
- C More water is absorbed by the root hairs when the stomata are closed.
- D More water is absorbed by the root hairs than is lost through the stomata.

12	B	1
----	---	---

34 The graph shows changes in the dry mass of a seed as it germinates and grows.



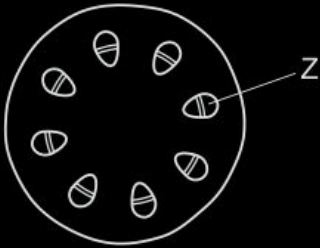
What causes the change shown between points S and T?

- A osmosis
- B photosynthesis
- C respiration
- D transpiration

35 The graph shows the concentration in the blood of three of the four hormones FSH, LH,

34	B	1
----	---	---

11 The diagram shows a section through a plant stem.



Which process is responsible for moving substances in region Z?

- A diffusion
- B osmosis
- C translocation
- D transpiration

MS 5090/12/O/N/21 Q11

11	C	1
----	---	---

5090/12/O/N/21 Q12

12 What is transpiration?

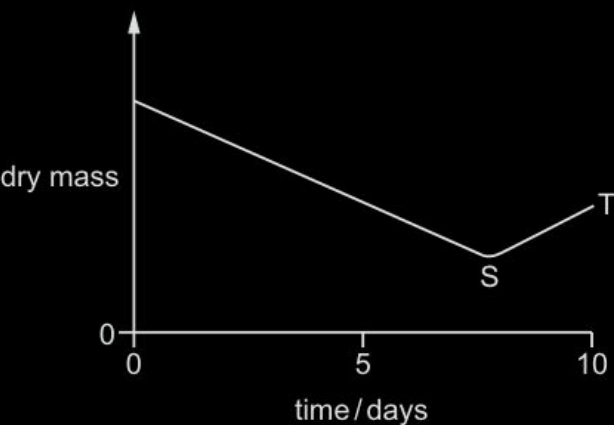
- A evaporation of water from the stomata
- B evaporation of water at the surface of mesophyll cells and its loss through the stomata
- C the movement of water through the cuticle
- D the movement of water in the xylem from the roots to the leaves

MS 5090/12/O/N/21 Q12

12	B	1
----	---	---

5090/12/O/N/21 Q34

34 The graph shows changes in the dry mass of a seed as it germinates and grows.



What causes the change shown between points S and T?

- A osmosis
- B photosynthesis
- C respiration
- D transpiration

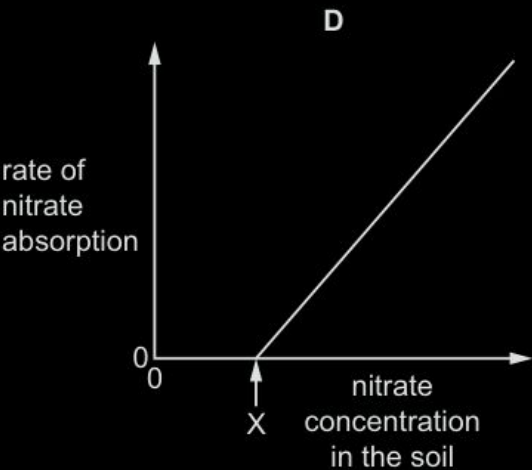
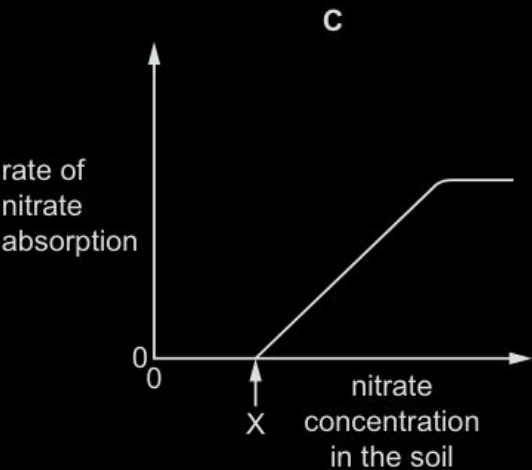
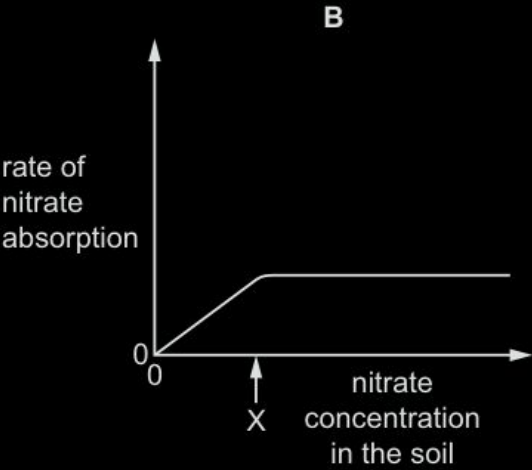
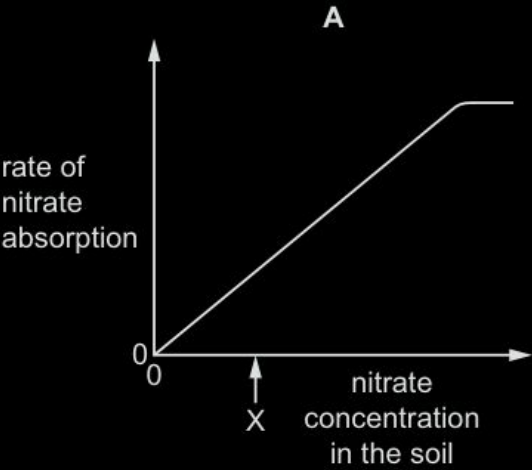
34	B	1
----	---	---

5090/11/M/J/22 Q4

4 The rate of nitrate ion absorption by a root hair cell was measured at different soil nitrate concentrations.

At X, the concentration of nitrate in the soil is the same as in the cell.

Which graph shows how the rate of absorption varies with nitrate concentration in the soil?



5 The graph shows how the rate of reaction of a protease from the stomach varies with pH.

4	A	1
---	---	---

13 What is the main cause of water moving up to the leaves in xylem vessels?

- A active transport
- B evaporation from the epidermis of the leaf
- C evaporation from the walls of the mesophyll cells
- D use of water in photosynthesis

14 Which blood vessels carry blood towards the heart and towards the lungs?

13	C	1
----	---	---

2 Xylem vessels are cells that have become adapted for conduction and support.

Which two adaptations assist them in these functions?

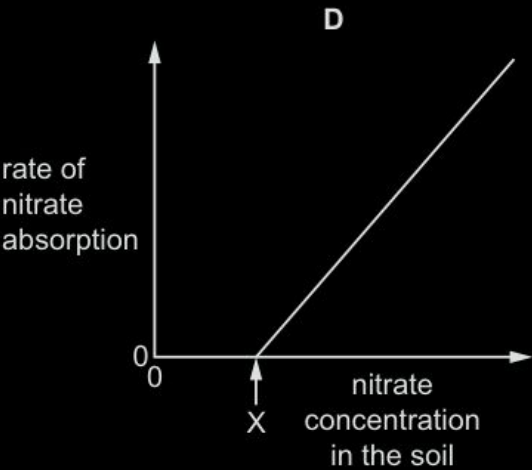
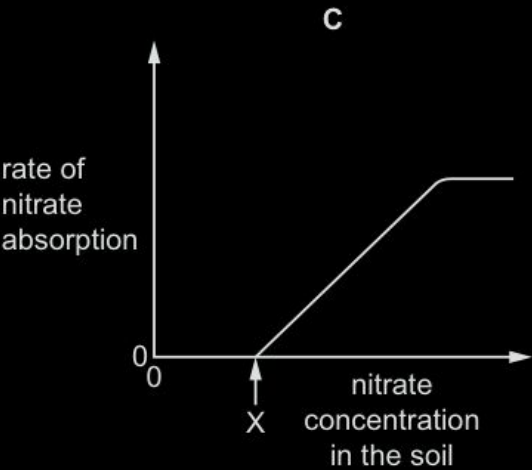
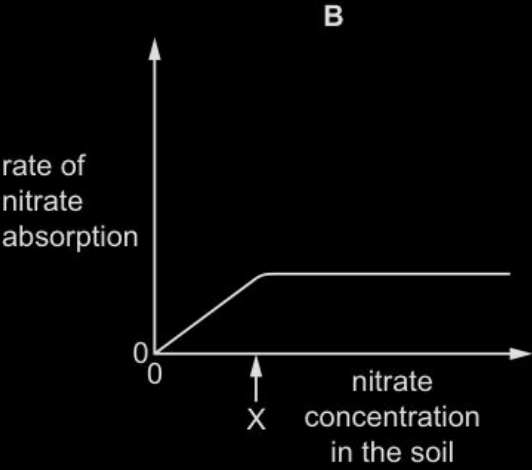
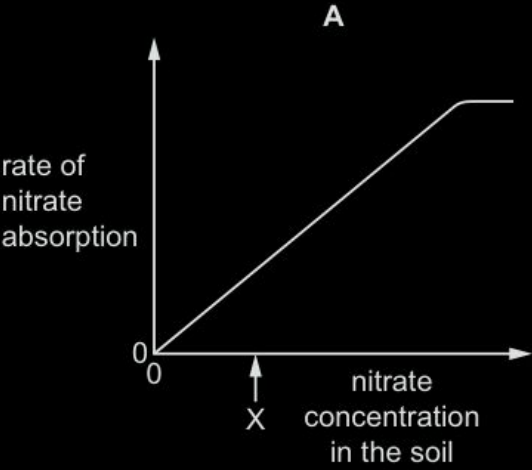
- A presence of a nucleus and cytoplasm
- B lack of cytoplasm and woody cell walls
- C lack of a nucleus and presence of cytoplasm
- D presence of cytoplasm and woody cell walls

2	B	1
---	---	---

4 The rate of nitrate ion absorption by a root hair cell was measured at different soil nitrate concentrations.

At X, the concentration of nitrate in the soil is the same as in the cell.

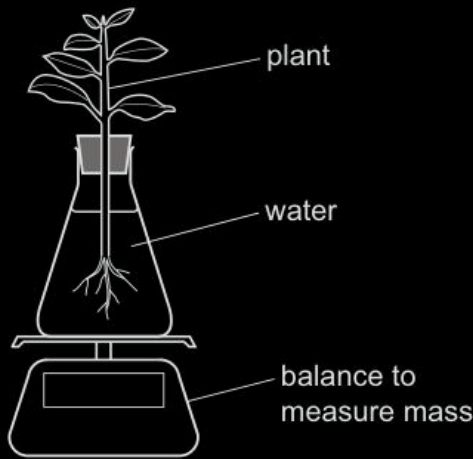
Which graph shows how the rate of absorption varies with nitrate concentration in the soil?



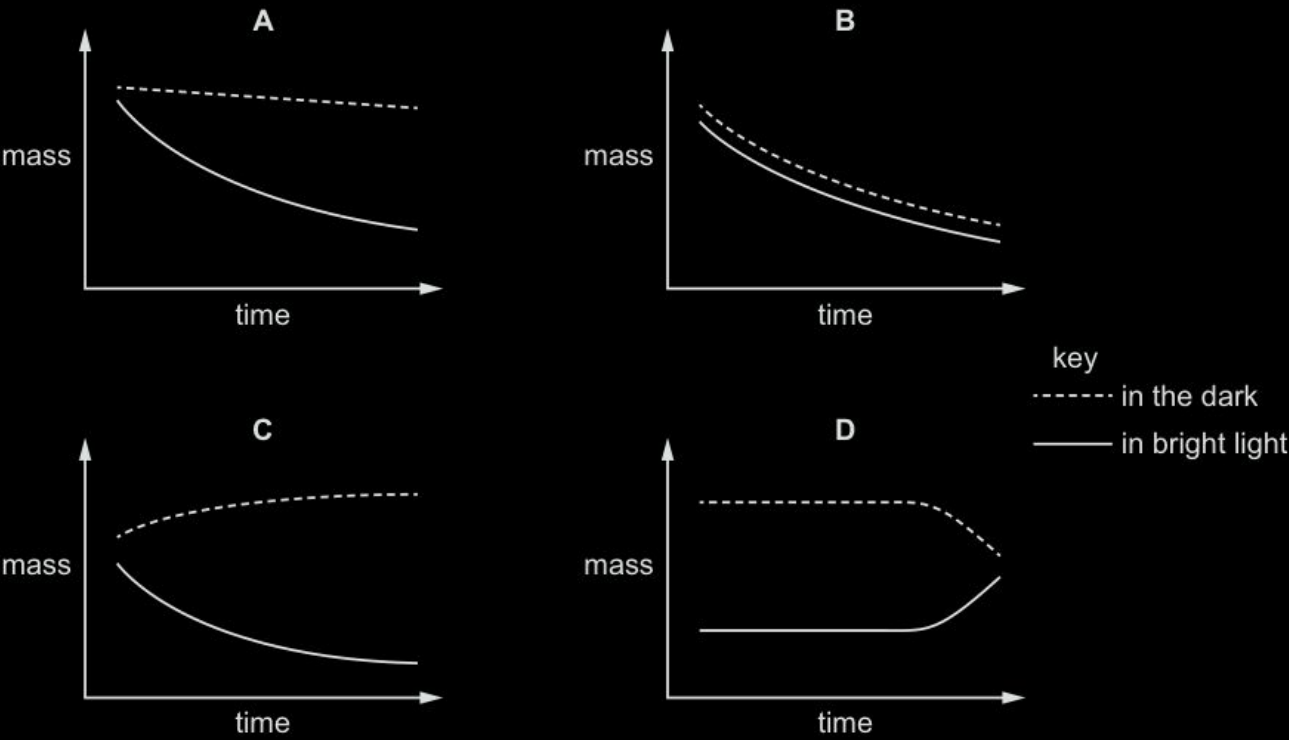
5 Amylase is an enzyme that breaks down starch to maltose.

4	A	1
---	---	---

12 A student set up an experiment to investigate the rate of transpiration in a plant.



The student set up two identical sets of apparatus.
One was placed in a room at 20 °C with bright light and one in a room at 20 °C in the dark.
Which graph shows the student's results?



MS 5090/12/M/J/22 Q12

12	A	1
----	---	---

5090/12/M/J/22 Q13

13 What is the main cause of water moving up to the leaves in xylem vessels?

- A active transport
- B evaporation from the epidermis of the leaf
- C evaporation from the walls of the mesophyll cells
- D use of water in photosynthesis

14 Which blood vessels, linked directly to the heart, transport deoxygenated blood?

13	C	1
----	---	---

2 What are the functions of xylem vessels?

	strengthening roots and stems	supporting leaf shape	transporting sugars
A	✓	x	x
B	✓	✓	x
C	x	✓	✓
D	x	x	✓

key
✓ = yes
x = no

2	B	1
---	---	---

4 Which row applies to active transport?

	movement occurs from region of higher concentration to region of lower concentration	movement occurs from region of lower concentration to region of higher concentration	an example is ion uptake by root hairs	an example is water uptake by root hairs
A	✓	x	✓	x
B	✓	x	x	✓
C	x	✓	✓	x
D	x	✓	x	✓

key
✓ = yes
x = no

4	C	1
---	---	---

11 Which function is **not** carried out by the liver?

- A breakdown of alcohol
- B conversion of glycogen to glucose
- C formation of urea
- D secretion of digestive enzymes

11	D	1
----	---	---

12 Which statements about root hairs are correct?

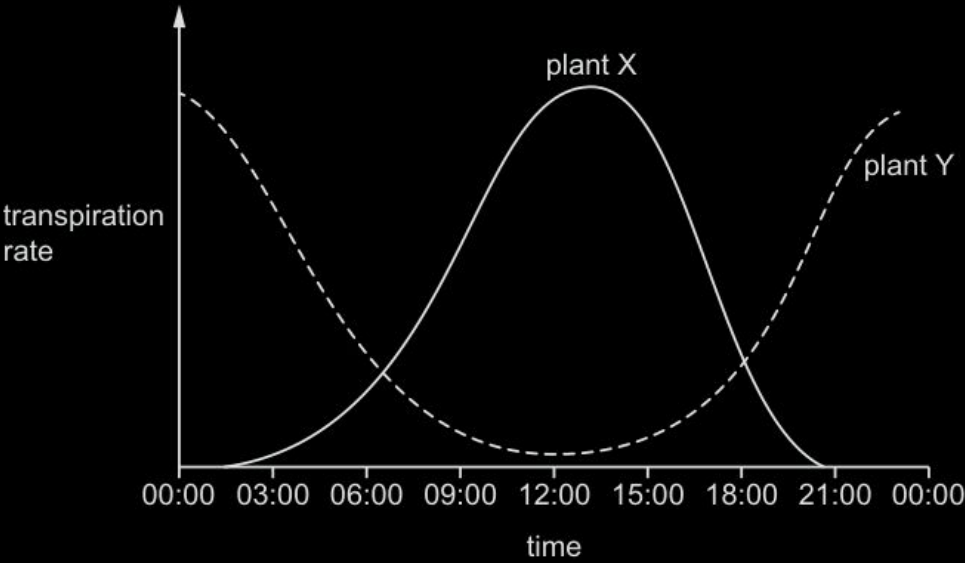
- 1 The cell membrane can allow movement of water and ions into the cell.
- 2 The cell vacuole extends into the root hair.
- 3 They are formed as an extension of the outer layer of root cells.

A 1, 2 and 3 B 1 and 2 only C 1 and 3 only D 2 and 3 only

13 The graph shows the transpiration rates of two plants during one day. Both plants were grown under identical conditions.

12	A	1
----	---	---

13 The graph shows the transpiration rates of two plants during one day. Both plants were grown under identical conditions.



Which statement explains what the graph shows?

- A Plant X closes its stomata during the brightest part of the day.
- B Plant Y closes its stomata during the brightest part of the day.
- C Plants X and Y close their stomata during the brightest part of the day.
- D Plant Y has no stomata.

13	B	1
----	---	---

20 How do the concentrations of oxygen and urea in the renal artery compare to their concentrations in the renal vein?

	oxygen in renal artery	urea in renal artery
A	higher	higher
B	higher	lower
C	lower	higher
D	lower	lower

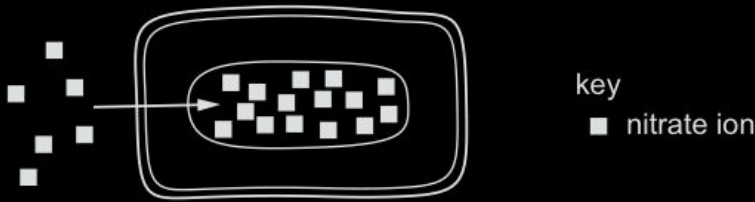
21 Which changes occur when a person walks from a very cold room into a hot room?

MS 5090/11/O/N/22 Q20

20	A	1
----	---	---

5090/12/O/N/22 Q4

4 Which process is moving nitrate ions into the cell shown?



- A active transport
- B diffusion
- C osmosis
- D translocation

MS 5090/12/O/N/22 Q4

4	A	1
---	---	---

5090/12/O/N/22 Q11

11 Which function is **not** carried out by the liver?

- A breakdown of alcohol
- B conversion of glycogen to glucose
- C formation of urea
- D secretion of digestive enzymes

12 Which statements about root hairs are correct?

MS 5090/12/O/N/22 Q11

11	D	1
----	---	---

12 Which statements about root hairs are correct?

- 1 The cell membrane can allow movement of water and ions into the cell.
- 2 The cell vacuole extends into the root hair.
- 3 They are formed as an extension of the outer layer of root cells.

A 1, 2 and 3 B 1 and 2 only C 1 and 3 only D 2 and 3 only

MS 5090/12/O/N/22 Q12

12	A	1
----	---	---

5090/12/O/N/22 Q13

13 The table shows weather conditions in a field.

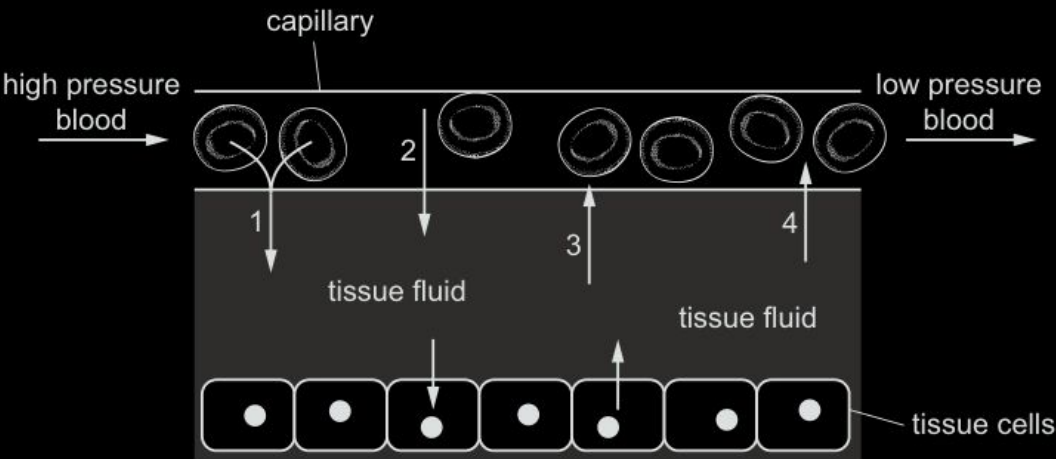
Which row shows conditions that would cause a plant to wilt most rapidly?

	air temperature /°C	humidity	time of day	wind speed /km per hour
A	15	raining	00:00	32
B	12	dry	14:00	24
C	6	fog / rain	08:00	3
D	5	dry	22:00	16

MS 5090/12/O/N/22 Q13

13	B	1
----	---	---

16 The diagram shows the transfer of materials between blood in a capillary and tissue fluid.



Which row shows the correct labels for the arrows in the diagram?

	1	2	3	4
A	carbon dioxide	glucose	oxygen	water
B	glucose	water	oxygen	urea
C	oxygen	glucose	carbon dioxide	water
D	oxygen	water	carbon dioxide	glucose

17 An experiment is set up, as shown.

16	C	1
----	---	---

20 What are two main functions of the kidneys?

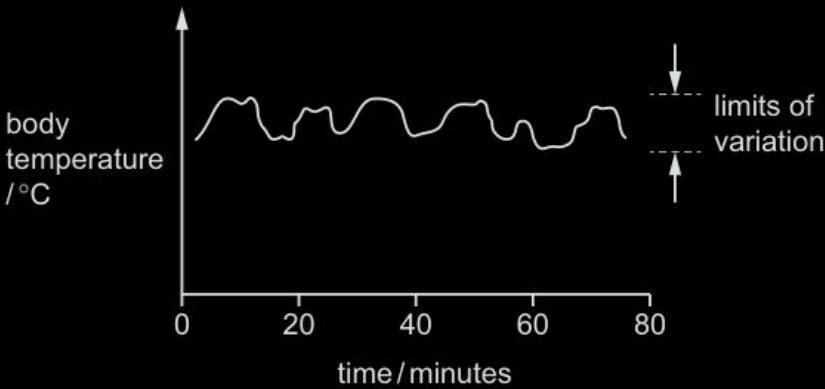
- A to make toxins in the blood harmless and to remove excess salt in the blood
- B to remove excess salt in the blood and to remove urea from the blood plasma
- C to remove urea from the blood plasma and to convert glucose to glycogen
- D to convert glucose to glycogen and to make toxins in the blood harmless

MS 5090/12/O/N/22 Q20

20	B	1
----	---	---

5090/12/O/N/22 Q22

22 The graph shows measurements taken of one aspect of the internal environment of the human body over time.



What controls the internal environment of the human body to keep each aspect within strict limits?

- A excretion
- B homeostasis
- C positive feedback
- D respiration

MS 5090/12/O/N/22 Q22

22	B	1
----	---	---

5090/11/M/J/23 Q9

9 Which description of the function of root hairs is correct?

- A Root hairs absorb ions by active transport and osmosis.
- B Root hairs absorb ions by osmosis only.
- C Root hairs absorb water by active transport.
- D Root hairs absorb water by osmosis.

MS 5090/11/M/J/23 Q9

9	D	1
---	---	---

10 Which conditions result in the highest rate of transpiration from a plant?

	percentage humidity	temperature / °C
A	60	15
B	60	25
C	100	15
D	100	25

11 Which disease can be the result of a dietary deficiency of both a mineral ion and a vitamin?

MS 5090/11/M/J/23 Q10

10	B	1
----	---	---

5090/11/M/J/23 Q12

12 A student states: 'The contraction of circular muscles pushes the contents forward.'

Which process is being described?

- A peristalsis
- B deamination
- C translocation
- D transpiration

MS 5090/11/M/J/23 Q12

12	A	1
----	---	---

5090/11/M/J/23 Q23

23 The blood in the renal vein has a different composition to the blood in the renal artery.

How is the blood in the renal vein different?

- A less carbon dioxide, less urea, less water
- B less oxygen, more urea, less water
- C more carbon dioxide, less urea, less water
- D more oxygen, less urea, less water

MS 5090/11/M/J/23 Q23

23	C	1
----	---	---

9 Root hairs take in water and ions from the soil.

Which table shows the correct concentrations of water and ions in the root hair?

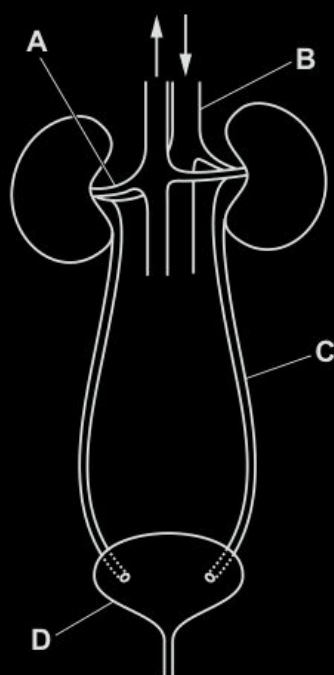
A		B	
substance	concentration in root hair	substance	concentration in root hair
water	lower than the soil	water	higher than the soil
ions	higher than the soil	ions	higher than the soil

C		D	
substance	concentration in root hair	substance	concentration in root hair
water	higher than the soil	water	lower than the soil
ions	lower than the soil	ions	lower than the soil

10 A herbaceous plant, growing in a nutrient solution, is placed in a well-lit container. Humid air is

9	A	1
---	---	---

23 In which structure is there a fluid with a low concentration of urea?

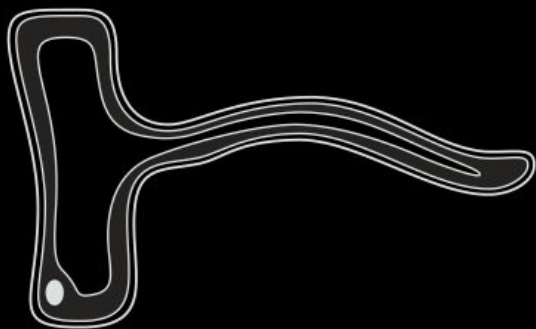


MS 5090/12/M/J/23 Q23

23	A	1
----	---	---

5090/11/O/N/23 Q9

9 The diagram shows a root hair cell.



How does the shape of the cell help the plant?

- A It allows roots to search for ions.
- B It allows roots to search for water.
- C It increases the surface area for sugar uptake.
- D It increases the surface area for water uptake.

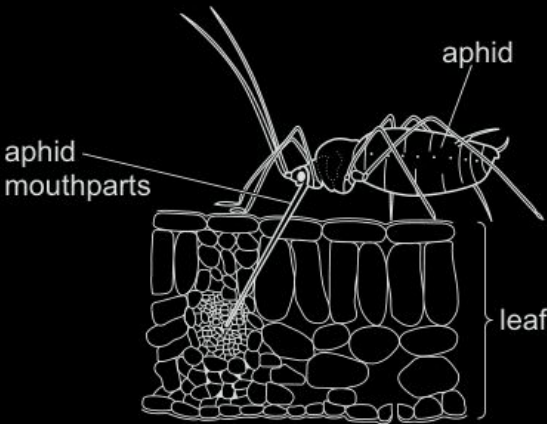
10 A student used the apparatus shown to measure the rate of water loss by the leafy stem at

MS 5090/11/O/N/23 Q9		
9	D	1

- 5090/12/O/N/23 Q1
- 1
- Which structures have both cytoplasm and cell walls?
- A liver cells
 - B red blood cells
 - C root hair cells
 - D xylem vessels

MS 5090/12/O/N/23 Q1		
1	C	1

- 5090/12/O/N/23 Q7
- 7
- The diagram shows an aphid feeding on sugar in a leaf.



When the aphid feeds, in which order do its mouthparts pass through the tissues of the leaf?

- A palisade cells → phloem → upper epidermis
- B upper epidermis → palisade cells → phloem
- C palisade cells → upper epidermis → xylem
- D upper epidermis → xylem → palisade cells

MS 5090/12/O/N/23 Q7		
7	B	1

- 5090/12/O/N/23 Q9
- 9
- Which row shows the processes involved when nitrate ions and water, from soil, enter a root hair cell?

	nitrate ions	water
A	active transport	osmosis
B	diffusion	osmosis
C	diffusion	active transport
D	osmosis	diffusion

- 10
- Which change in conditions will decrease the rate of transpiration?

MS 5090/12/O/N/23 Q9		
9	A	1

10 Which change in conditions will decrease the rate of transpiration?

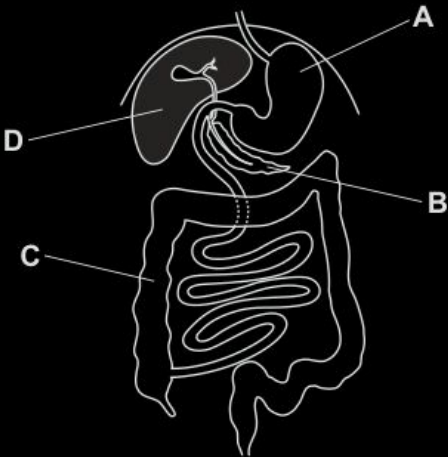
- A a decrease in air currents around the leaf
- B a decrease in the humidity of the atmosphere
- C an increase in light intensity around the leaf
- D an increase in the temperature of the atmosphere

MS 5090/12/O/N/23 Q10

10	A	1
----	---	---

5090/12/O/N/23 Q23

23 Which label shows the location for the production of urea?

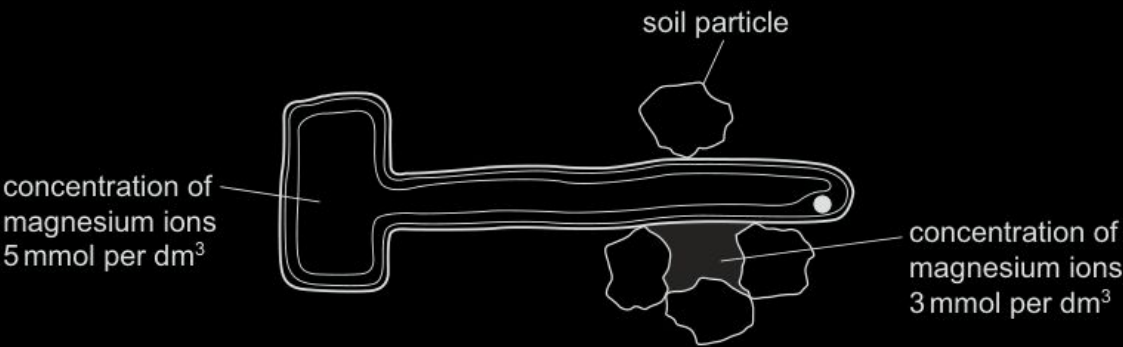


24 A student was asked to compare a typical spinal reflex action with a voluntary action.

MS 5090/12/O/N/23 Q23

23	D	1
----	---	---

4 The diagram shows the concentration of magnesium ions in a root hair cell of a healthy plant and in the soil water surrounding the root hair cell. The plant continuously uses up magnesium ions.



For the plant to remain healthy, how will the magnesium ions move?

- A into the cell by active transport
- B into the cell by diffusion
- C out of the cell by active transport
- D out of the cell by diffusion

MS 5090/11/M/J/24 Q4

4	A	1
---	---	---

5090/11/M/J/24 Q9

9 Which pathway is taken by water and mineral ions through a plant?

- A root hair cells → mesophyll cells → xylem → cortex cells
- B root hair cells → cortex cells → phloem → mesophyll cells
- C root hair cells → mesophyll cells → phloem → cortex cells
- D root hair cells → cortex cells → xylem → mesophyll cells

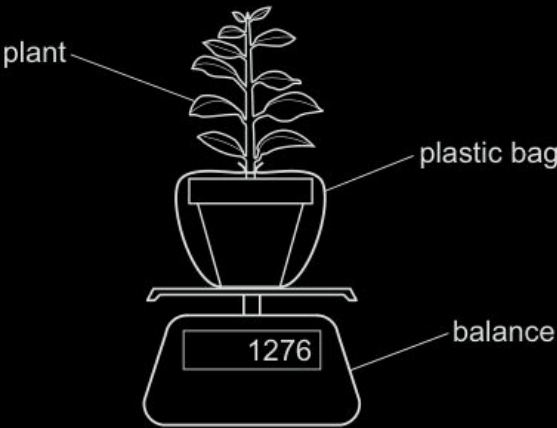
10 A student investigated the rate of transpiration in a green plant. The plant was growing in soil in

MS 5090/11/M/J/24 Q9

9	D	1
---	---	---

10 A student investigated the rate of transpiration in a green plant. The plant was growing in soil in a pot.

The student watered the plant and then put a plastic bag around the pot, leaving the leaves of the plant outside the bag. She measured the mass of the plant in the pot at the start and repeated this every two hours for six hours.



The results of the investigation are shown.

time of day	mass of pot and plant/g
09:00	1276
11:00	1270
13:00	1258
15:00	1252

What is the mean rate of transpiration over the six-hour period?

- A 43g/h
- B 12g/h
- C 8g/h
- D 4g/h

11 The diagram shows what happens to amino acids in part of the human body.

10	D	1
----	---	---

13 A healthy person eats a meal containing carbohydrates.

Two hours later, which structure will contain a liquid with an increased concentration of glucose?

- A colon
- B hepatic portal vein
- C renal artery
- D ureter

MS 5090/11/M/J/24 Q13

13	B	1
----	---	---

5090/11/M/J/24 Q22

22 Which row identifies the contents of urine in a healthy person?

	glucose	urea	water	ions
A	✓	x	✓	x
B	✓	✓	✓	✓
C	x	x	✓	✓
D	x	✓	✓	✓

MS 5090/11/M/J/24 Q22

22	D	1
----	---	---

5090/11/M/J/24 Q26

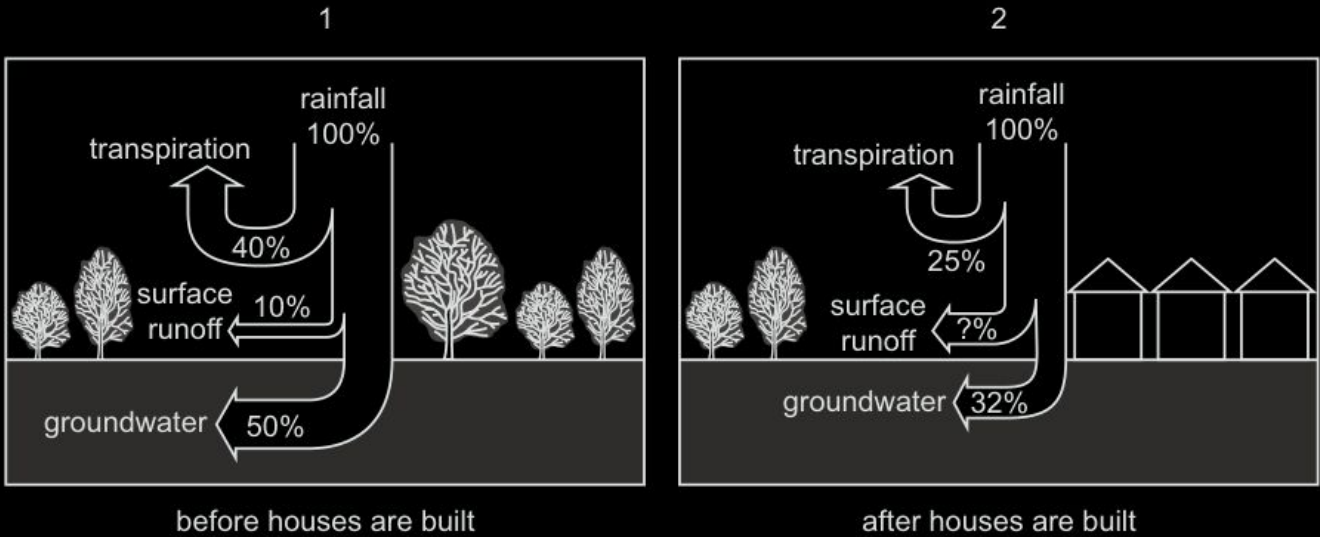
26 Which process is stimulated by adrenaline in the cells of the liver?

- A breakdown of glycogen, increasing the blood glucose level
- B breakdown of excess amino acids, forming urea
- C breakdown of proteins, releasing amino acids into the blood
- D conversion of excess blood glucose to glycogen

MS 5090/11/M/J/24 Q26

26	A	1
----	---	---

39 The diagrams show what happens to rain when it falls on an area of forested land before and after houses are built.



What is the difference in the percentage of surface runoff before and after building the houses?

- A increases by 33%
- B increases by 43%
- C decreases to 43%
- D decreases to 33%

39	A	1
----	---	---

9 Water moves from the soil to the atmosphere through a plant.

The water enters the plant through the root hair cells and moves into the root cortex.

Where does the water move into from the root cortex?

- A into the mesophyll cells
- B into the phloem cells
- C into the xylem vessels
- D through the stomata

10 The graph shows the loss of water vapour from two different plants growing in identical

MS 5090/12/M/J/24 Q9

9	C	1
---	---	---

5090/12/M/J/24 Q21

21 What is an example of excretion?

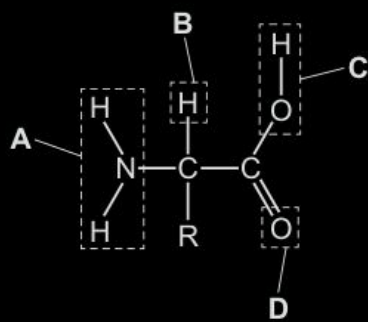
- A release of a hormone into the blood
- B removal of carbon dioxide from the lungs
- C removal of undigested food from the digestive system
- D release of water from the sweat glands

MS 5090/12/M/J/24 Q21

21	B	1
----	---	---

22 The diagram shows an amino acid molecule.

Which part of the amino acid molecule is removed and used to make urea?



23 What is a function of sensory neurones?

MS 5090/12/M/J/24 Q22

22	A	1
----	---	---

5090/12/M/J/24 Q38

38 Which process is involved in the conversion of ammonia to nitrates?

- A denitrification
- B excretion
- C nitrification
- D nitrogen fixation

39 Cutting down tropical rainforest trees has many consequences.

MS 5090/12/M/J/24 Q38

38	C	1
----	---	---

5090/11/O/N/24 Q4

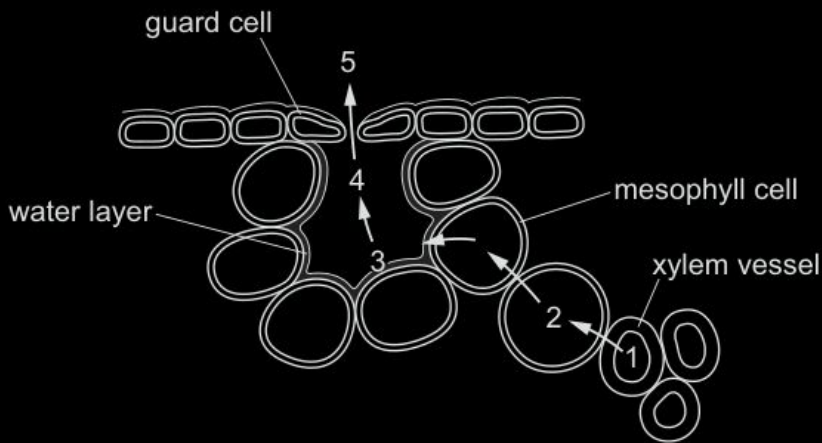
4 Which movement of a substance in a plant requires the plant to provide energy?

- A absorption of carbon dioxide by a palisade cell
- B absorption of oxygen by a mesophyll cell
- C nitrate uptake by root hair cells
- D transport of water up through the xylem

MS 5090/11/O/N/24 Q4

4	C	1
---	---	---

10 The diagram shows the movement of water through part of a leaf.



Which processes are involved in the movement of water at these stages?

	1–2	3–4	4–5
A	diffusion	evaporation	osmosis
B	evaporation	diffusion	osmosis
C	osmosis	diffusion	evaporation
D	osmosis	evaporation	diffusion

11 The diagram shows part of the human digestive svstem.

MS 5090/11/O/N/24 Q10

10	D	1
----	---	---

5090/11/O/N/24 Q15

15 Scientists have concluded that absorpion of mineral ions in plants requires energy from respiration.

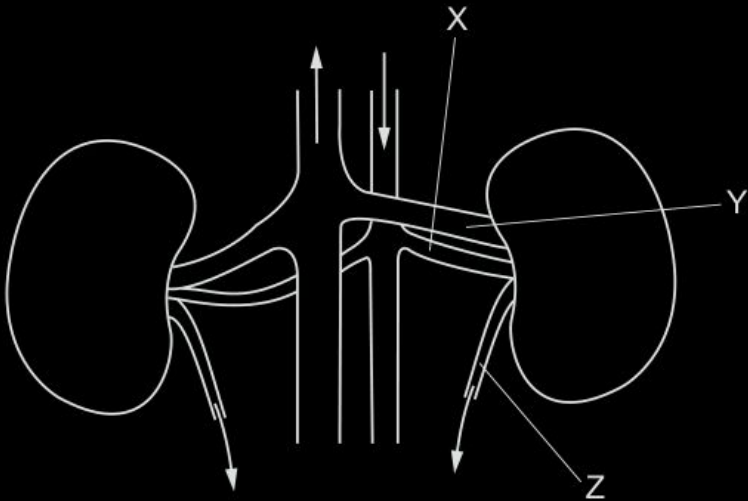
Which observation best supports this conclusion?

- A Carbohydrate is stored in the roots.
- B Living roots give off carbon dioxide.
- C The root hairs have a large surface area.
- D Uptake of mineral ions is reduced in lower oxygen concentrations.

MS 5090/11/O/N/24 Q15

15	D	1
----	---	---

22 The diagram shows part of the human urinary system.

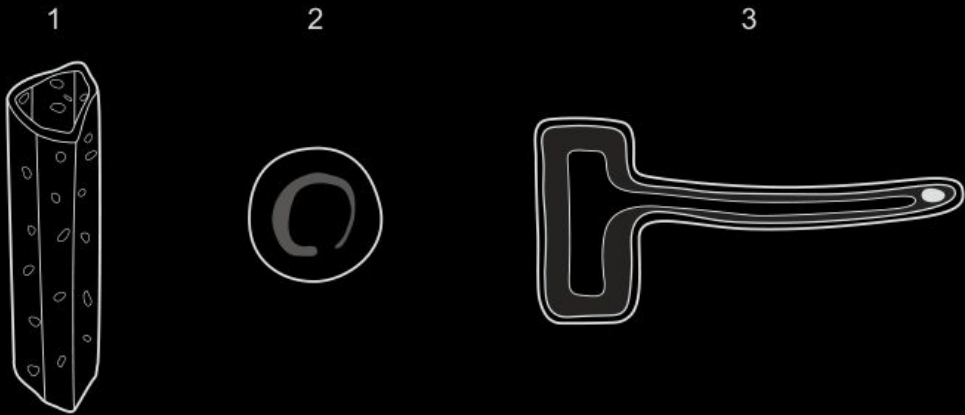


What are structures X, Y and Z?

	X	Y	Z
A	artery	vein	ureter
B	artery	vein	urethra
C	vein	artery	ureter
D	vein	artery	urethra

22	A	1
----	---	---

1 The diagram shows three different types of microscopic structure.



What are these structures?

	1	2	3
A	xylem vessel	root hair cell	red blood cell
B	red blood cell	xylem vessel	root hair cell
C	root hair cell	red blood cell	xylem vessel
D	xylem vessel	red blood cell	root hair cell

1	D	1
---	---	---

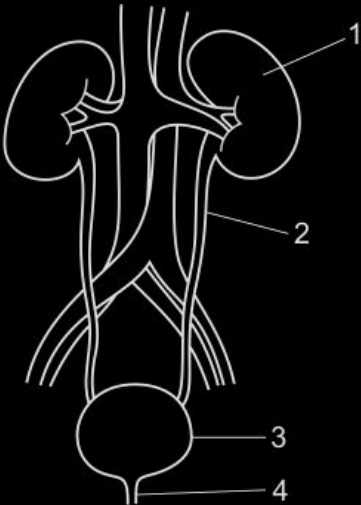
9 Root hair cells take in water and ions from the soil.

Which row shows how water and ions are taken into root hair cells?

	water	ions
A	active transport	active transport
B	active transport	osmosis
C	osmosis	osmosis
D	osmosis	active transport

9	D	1
---	---	---

22 The diagram shows the excretory system.

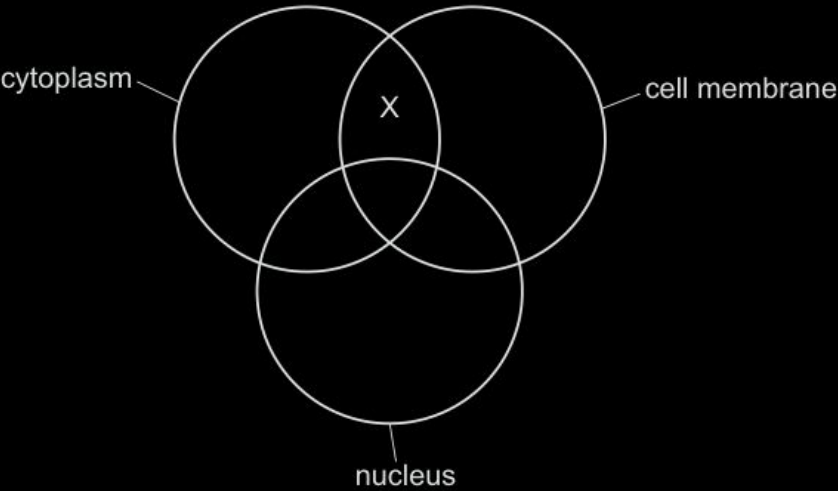


Which row identifies the names of the labelled structures?

	1	2	3	4
A	bladder	ureter	kidney	urethra
B	bladder	urethra	kidney	ureter
C	kidney	ureter	bladder	urethra
D	kidney	urethra	bladder	ureter

22	C	1
----	---	---

3 The diagram shows the structures found in cell X.

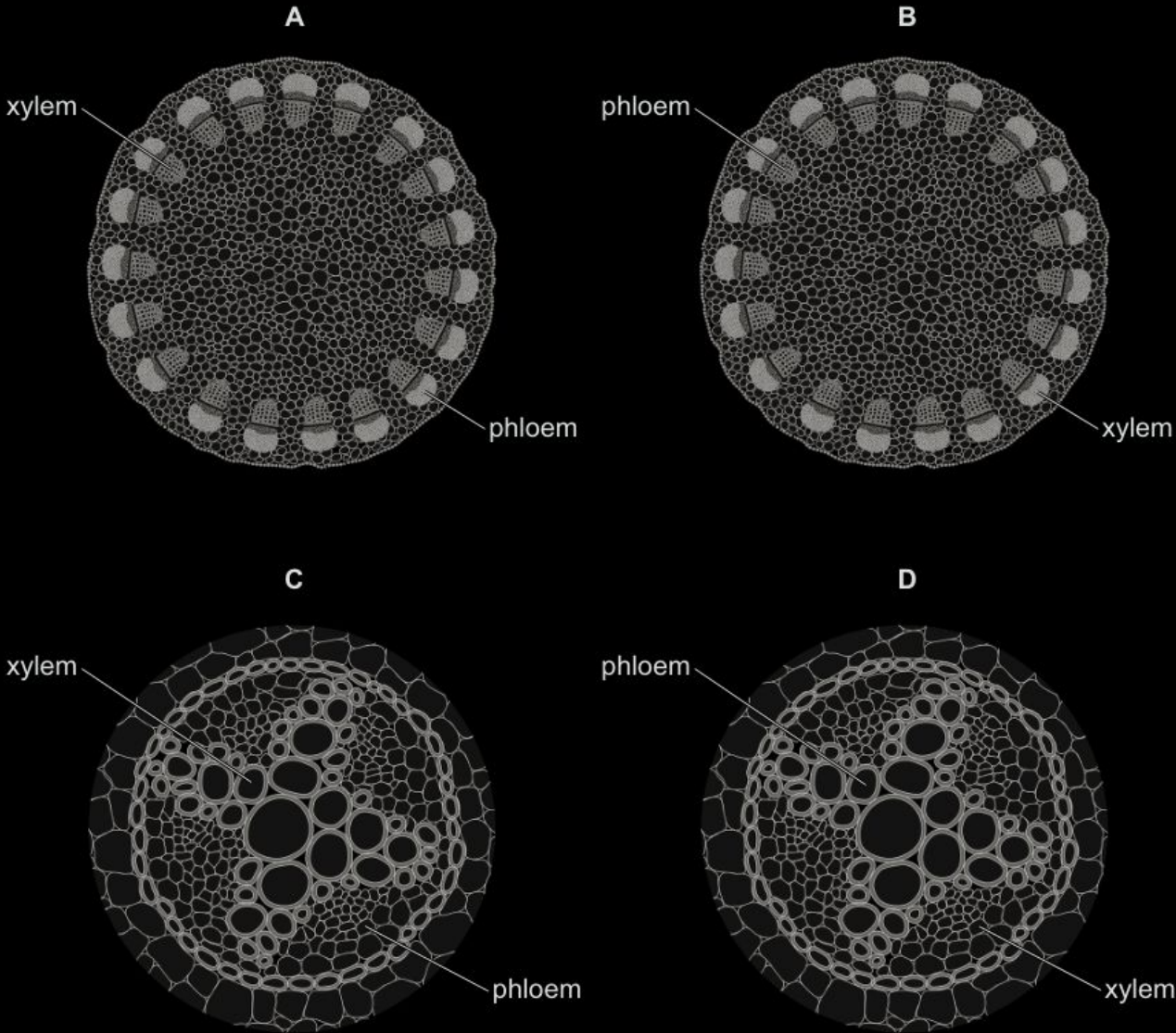


Which type of cell is X?

- A liver cell
- B red blood cell
- C root hair cell
- D xylem vessel

3	B	1
---	---	---

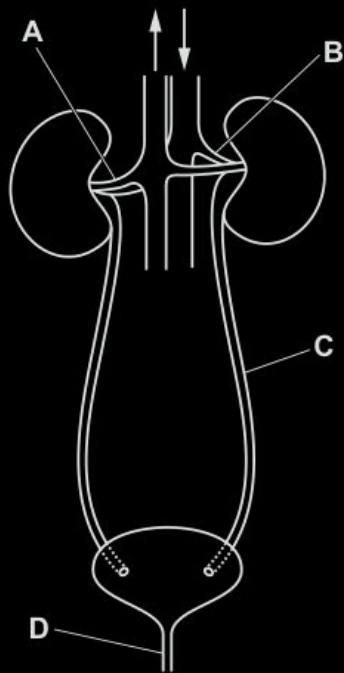
12 Which transverse section photomicrograph shows the position of the xylem and phloem in a non-woody dicotyledonous stem?



13 Students set up apparatus to measure the uptake of water by a leafy shoot.

12	A	1
----	---	---

23 Which structure is the ureter?



24 The processes of filtration and reabsorption take place in the kidneys.

MS 5090/11/M/J/25 Q23

23	C	1
----	---	---

24 The processes of filtration and reabsorption take place in the kidneys.

The daily amounts of some of the substances entering the kidneys in the blood and of these substances leaving the kidneys in the urine are shown.

substance	amount entering the kidneys in the blood	amount leaving the kidneys in the urine
water	150 dm ³	1.5 dm ³
glucose	150 g	0 g
urea	50 g	30 g

Which percentage of each substance is reabsorbed by the kidneys during a day?

	% of substance reabsorbed by kidneys		
	water	glucose	urea
A	99	100	40
B	148.5	150	20
C	0.99	1.0	0.4
D	1.0	0.0	60

MS 5090/11/M/J/25 Q24

24	A	1
----	---	---

5090/11/M/J/25 Q29

29 A nurse tests a sample of urine from a patient using Benedict's reagent. The sample tests positive.

What is a possible explanation for this result?

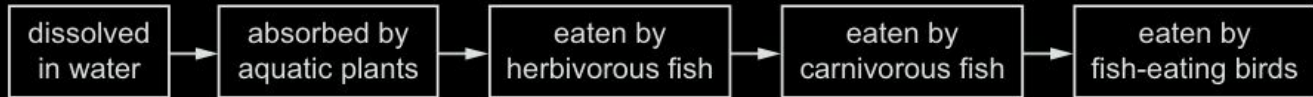
- A The patient's liver is **not** deaminating protein efficiently.
- B The patient's pancreas is **not** producing sufficient insulin.
- C The patient has eaten too much glucose.
- D The patient's concentration of plasma proteins is too high.

MS 5090/11/M/J/25 Q29

29	B	1
----	---	---

- 40** DDT is an insecticide that is harmful to living organisms. It builds up in the cells of living organisms as it is **not** excreted.

The diagram shows the path taken by DDT when it enters a lake.



Which organisms would the DDT harm the most?

- A** aquatic plants
- B** carnivorous fish
- C** herbivorous fish
- D** fish-eating birds

40	D	1
----	---	---

5090/12/M/J/25 Q11

11 Glucose produced by photosynthesis can be converted into other carbohydrates or used immediately as glucose.

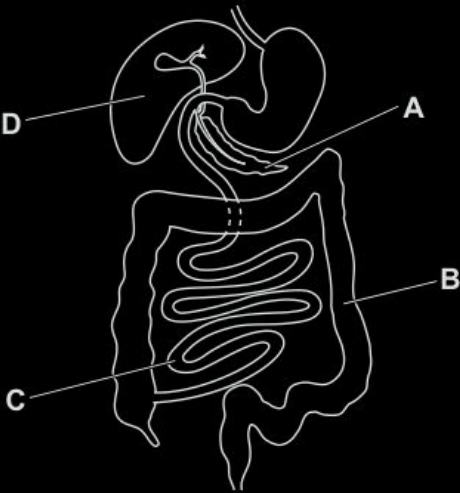
What is an immediate use of glucose in plants?

- A** building cell walls
- B** providing energy in respiration
- C** storing energy
- D** transporting energy in the phloem

MS 5090/12/M/J/25 Q11

11	B	1
----	---	---

25 Which letter shows where urea is produced?



MS 5090/12/M/J/25 Q25

25	D	1
----	---	---

5090/12/M/J/25 Q26

26 Which substance would **not** be present in the urine of a healthy person with a normal diet?

- A glucose
- B salts
- C urea
- D water

27 Which part of the eye contains the greatest density of light receptors?

MS 5090/12/M/J/25 Q26

26	A	1
----	---	---

5090/11/O/N/25 Q5

5 Which process involves the movement of molecules against a concentration gradient?

- A active transport
- B diffusion
- C osmosis
- D transpiration

MS 5090/11/O/N/25 Q5

5	A	1
---	---	---

14 Which processes use energy obtained from respiration?

- 1 contraction of intercostal muscles during breathing
- 2 nitrate ions entering a root hair cell against the concentration gradient
- 3 oxygen moving from the lungs into blood capillaries
- 4 production of antibodies by lymphocytes
- 5 water entering a bacterial cell by osmosis

A 1, 2 and 3 B 1, 2 and 4 C 2, 3 and 4 D 3, 4 and 5

MS 5090/11/O/N/25 Q14

14	B	1
----	---	---

5090/11/O/N/25 Q15

15 During vigorous exercise, lactic acid builds up in the muscles. Blood leaving these muscles will have a high concentration of lactic acid.

Which process removes **most** of the lactic acid from the blood?

- A It is broken down in the liver.
- B It is excreted during sweating.
- C It is expired during faster breathing.
- D It is oxidised to release energy in muscles.

16 The table shows the mean rate of blood flow through some blood vessels.

MS 5090/11/O/N/25 Q15

15	A	1
----	---	---

21 The table shows the concentration of five substances in the blood entering the kidney, in fluid entering the nephron and in the urine.

substance	concentration / mg per dm ³		
	blood entering the kidney	fluid entering the nephron	urine
urea	0.4	20.0	20.0
glucose	1.5	1.5	0.0
amino acids	0.8	0.8	0.0
ions	8.0	8.0	16.5
protein	82.0	0.0	0.0

Which substances are totally reabsorbed into the blood from the nephron?

- A amino acids and glucose
- B glucose and ions
- C protein and amino acids
- D protein and glucose

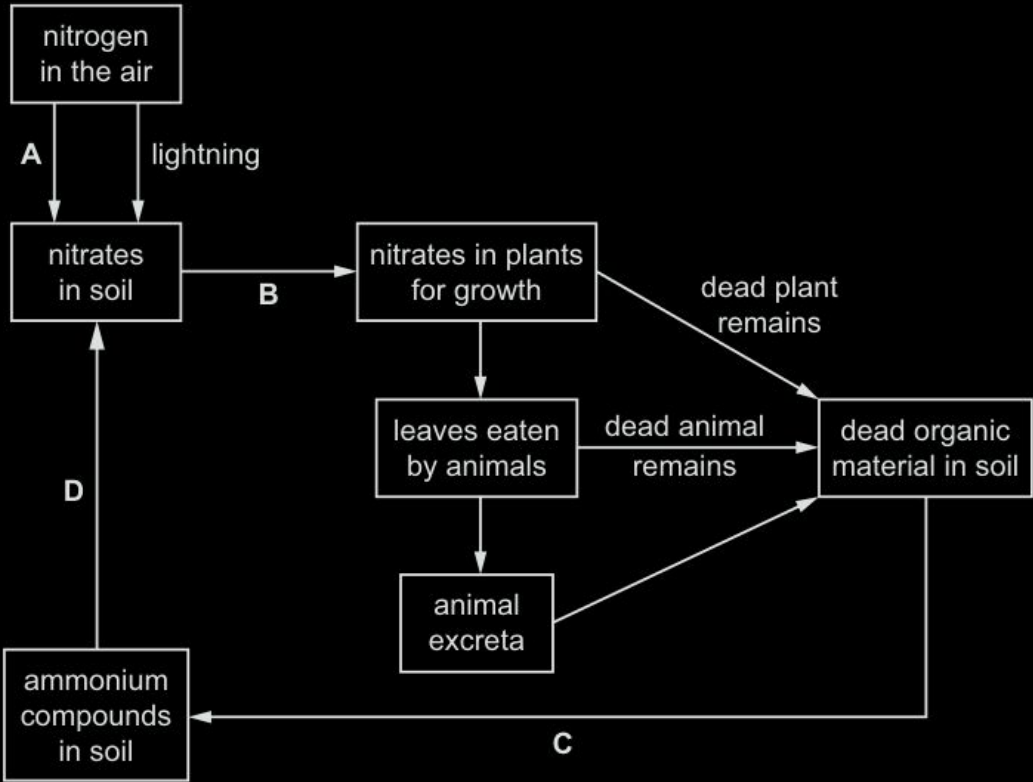
22 The graph shows blood glucose concentration over 6 hours in a day.

21	A	1
----	---	---

39 Trees in a forest need a constant supply of nitrates to continue growing.

The diagram shows some of the stages of the nitrogen cycle that provide the necessary nitrates.

Which stage in this cycle is **not** brought about by living bacteria?



MS 5090/11/O/N/25 Q39

39	B	1
----	---	---

5090/12/O/N/25 Q8

8 Oxygen is produced in leaf cells during photosynthesis and diffuses out of the leaf.

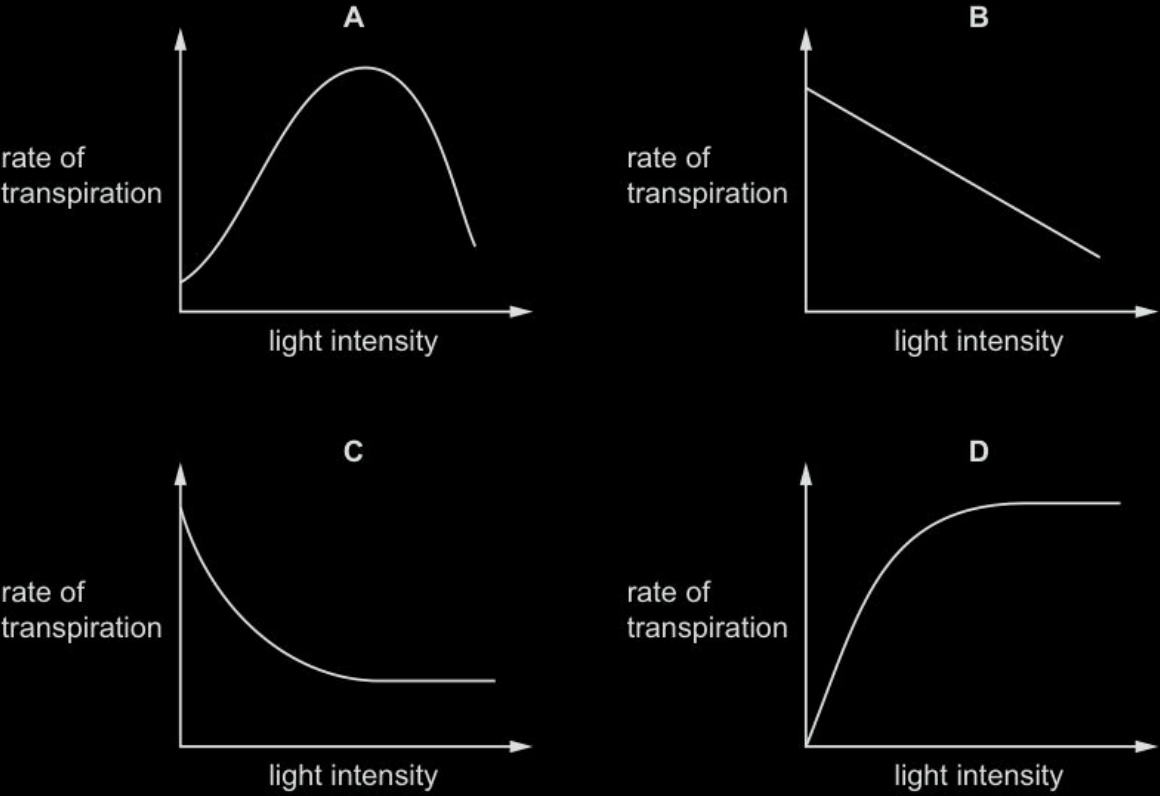
By which pathway does the oxygen diffuse?

- A through the upper epidermis and through the cuticle
- B through the airspaces and through the stomata
- C through the mesophyll and through the phloem
- D through the mesophyll and through the xylem

MS 5090/12/O/N/25 Q8

8	B	1
---	---	---

10 Which graph shows the effect of light intensity on the rate of transpiration?

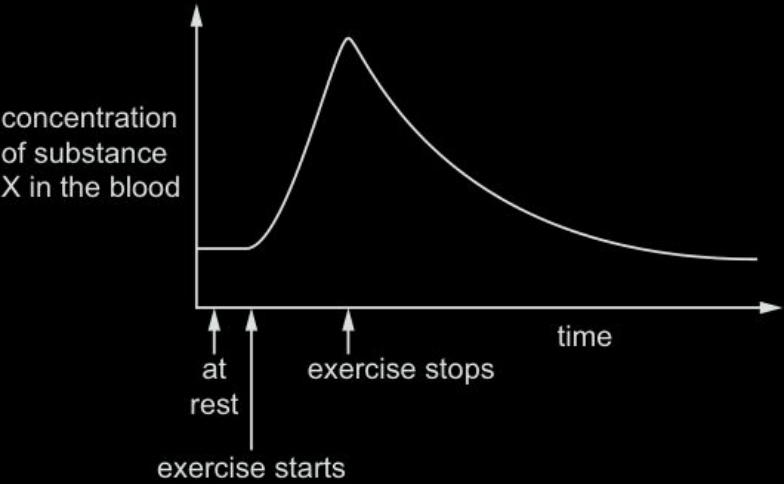


MS 5090/12/O/N/25 Q10

10	D	1
----	---	---

5090/12/O/N/25 Q14

14 The graph shows the concentration of substance X in a person's blood before, during and after exercise.



What is substance X?

- A alcohol
- B glycogen
- C lactic acid
- D urea

MS 5090/12/O/N/25 Q14

14	C	1
----	---	---

23 Which term can be used to describe the release of oxygen from plants?

- A breathing
- B excretion
- C respiration
- D transpiration

24 The diagram shows some structures in the skin.

MS 5090/12/O/N/25 Q23

23	B	1
----	---	---

5090/12/O/N/25 Q35

35 Bread is made from a mixture of flour, sugar, water and yeast.

The mixture is left in a warm place to allow it to rise before being baked.

Which process makes the bread rise?

- A The yeast feeds on the flour and grows.
- B The yeast reproduces to make the loaf bigger.
- C The yeast uses sugar to respire and produces carbon dioxide.
- D The yeast produces alcohol which it excretes.

MS 5090/12/O/N/25 Q35

35	C	1
----	---	---